Chapter 13: Reactions Equilibria in Systems Containing Components in Condensed Solution Part III

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Outlines

13.7 Graphical Representation Of Phase Equilibria

13.7.1 Phase Equilibria in the System Mg–Al–O

- ✓ Consider the phase equilibria in the system Mg–Al–O at 1073 K. At 1073 K, liquid Mg and Al are completely miscible in one another, and MgO, Al_2O_3, and the *spinel* * $MgAl_2O_4$ occur as the products of oxidation of the metallic alloys.
- ✓ The stabilities of the oxides are determined by the activities of Al and Mg in the liquid metallic alloys and by the standard Gibbs free energies of formation of the oxides.

* A spinel is a cubic oxide with the general formula AB_2O_4, where A is a divalent and B is a trivalent ion.

13.7 Graphical Representation Of Phase Equilibria

- ✓ For

$$2Al_{(l)} + \frac{3}{2}O_{2(g)} = Al_2O_{3(s)} \qquad \text{(i)}$$

$$\Delta G^\circ_{\text{(i)},1073\text{ K}} = -1,323,000 \text{ J} = -RT\ln\frac{a_{Al_2O_3}}{a_{Al}^2 p_{O_2}^{3/2}}$$

which gives

$$2\log a_{Al} + 1.5\log p_{O_2} - \log a_{Al_2O_3} = -64.39 \qquad \text{(ia)}$$

- ✓ For

$$2Mg_{(l)} + \frac{1}{2}O_{2(g)} = MgO_{(s)} \qquad \text{(ii)}$$

$$\Delta G^\circ_{\text{(ii)},1073\text{ K}} = -484,300 \text{ J} = -RT\ln\frac{a_{MgO}}{a_{Mg} p_{O_2}^{1/2}}$$

which gives

$$\log p_{O_2} = -2\log a_{Mg} + \log a_{MgO} - 47.14 \qquad \text{(iia)}$$

13.7 Graphical Representation Of Phase Equilibria

- ✓ For

$$\mathrm{Mg}_{(l)} + 2\mathrm{Al}_{(l)} + 2\mathrm{O}_{2(g)} = \mathrm{MgAl_2O}_{4(s)} \qquad \text{(iii)}$$

$$\Delta G^{\circ}_{\text{(iii)},1073\text{ K}} = -1,854,000\text{ J} = -RT\ln\frac{a_{\mathrm{MgAl_2O_4}}}{a_{\mathrm{Mg}}a_{\mathrm{Al}}^2 p_{\mathrm{O_2}}^2}$$

which gives

$$\log p_{\mathrm{O_2}} = \frac{1}{2}\log a_{\mathrm{Mg}} - \log a_{\mathrm{Al}} + \log a_{\mathrm{MgAl_2O_4}} - 90.24 \qquad \text{(iiia)}$$

- ✓ Combination of the reactions given by Equations (i), (ii), and (iii) gives

$$\mathrm{MgO}_{(s)} + \mathrm{Al_2O}_{3(s)} = \mathrm{MgAl_2O}_{4(s)} \qquad \text{(iv)}$$

for which

$$\Delta G^{\circ}_{\text{(iv)},1073\text{ K}} = -46,700\text{ J} = -RT\ln\frac{a_{\mathrm{MgAl_2O_4}}}{a_{\mathrm{MgO}}a_{\mathrm{Al_2O_3}}}$$

or

$$\log a_{\mathrm{MgO}} + \log a_{\mathrm{Al_2O_3}} - \log a_{\mathrm{MgAl_2O_4}} = -2.273 \qquad \text{(iva)}$$

13.7 Graphical Representation Of Phase Equilibria

- ✓ The Gibbs free energy of mixing diagram for the system $MgO–Al_2O_3$ is shown in Figure 13.17.
- ✓ From $\Delta G^{\circ}_{\text{(iv)},1073\ \text{K}}$, the Gibbs free energy of formation of $(MgO)_{1/2}\ (Al_2O_3)_{1/2}$ is –23,350 J, and the existence of $MgAl_2O_4$ as a stable phase requires that the activities of both MgO and Al_2O_3 have values between log *a* = –2.273 and log *a* = 0, the values of which are determined by Equation (iva).
- ✓ From Equation (iva) and Figure 13.17, the logarithm of the activity of Al_2O_3 in $MgAl_2O_4$ (at $a_{MgAl_2O_4} = 1$) which is saturated with MgO (at $a_{MgO} = 1$) is –2.273, and, from symmetry, the logarithm of the activity of MgO in $MgAl_2O_4$ (at $a_{MgAl_2O_4} = 1$) which is saturated with Al_2O_3 (at $a_{Al_2O_3} = 1$) is -2.273.
- ✓ Thus, if the activity of either MgO or Al_2O_3 in the system is less than antilog (– 2.273), the spinel of $MgAl_2O_4$ is not stable.

13.7 Graphical Representation Of Phase Equilibria

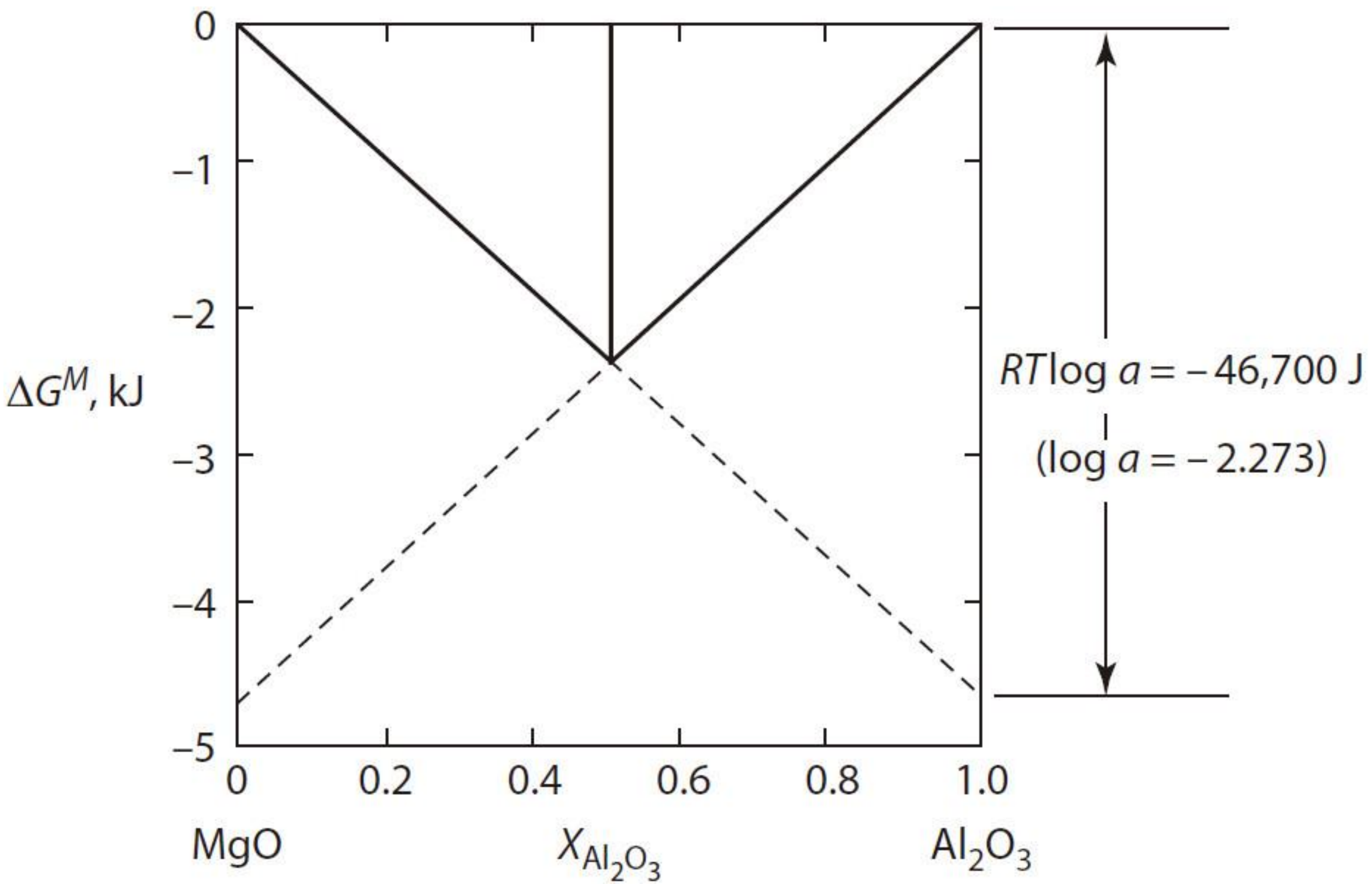

Figure 13.17 Molar Gibbs free energies of mixing in the system $MgO{-}Al_2O_3$ at 1073 K.

13.7 Graphical Representation Of Phase Equilibria

- ✓ The measured activities of Mg and Al at 1073 K in the system Mg–Al have been fitted by the equations

$$\log a_{\mathrm{Mg}} = \log X_{\mathrm{Mg}} - 0.68\left(1 - X_{\mathrm{Mg}}\right)^3$$

and

$$\log a_{\mathrm{Al}} = \log\left(1 - X_{\mathrm{Mg}}\right) - 1.02 X_{\mathrm{Mg}}^2 + 0.68 X_{\mathrm{Mg}}^3$$

- ✓ The phase stability diagram for the system at 1073 K, using p_{O_2} and $\log a_{\mathrm{Mg}}$ as coordinates, is shown in Figure 13.18a (the fixing of the activities of O_2 and Mg in the ternary system Mg–Al–O at constant temperature fixes the activity of Al).

13.7 Graphical Representation Of Phase Equilibria

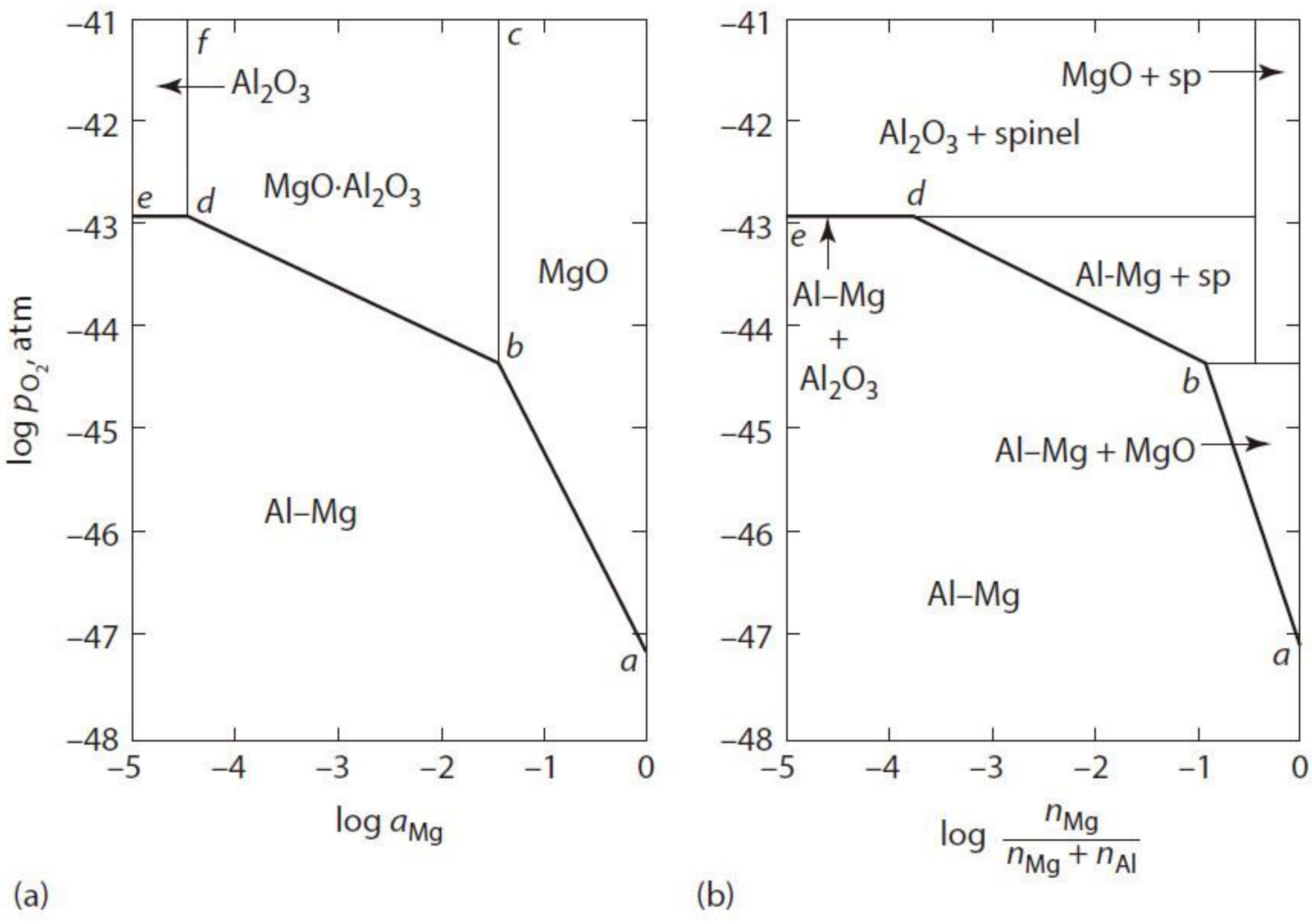

Figure 13.18 (a) The phase stability diagram for the system Al–Mg–O at 1073 K. (b) The phase diagram for the system Al–Mg–O at 1073 K.

13.7 Graphical Representation Of Phase Equilibria

- ✓ From Equation (iia), equilibrium between pure liquid Mg and MgO occurs at $\log p_{O_2} = -47.17$, which is shown as the point *a* in Figure 13.18a.
- ✓ The three-phase equilibrium involving the Mg–Al melt, MgO, and the gas phase is determined by Equation (iia) and is shown as the line *ab* in Figure 13.18a.
- ✓ The addition of Al to the liquid alloy decreases a_{Mg} and hence increases the value of the pressure of oxygen required to maintain $a_{\mathrm{MgO}} = 1$.
- ✓ Also, in moving along *ab* from *a* toward *b*, the activity of $a_{\mathrm{Al_2O_3}}$, given by Equation (i), increases, and at the point *b*, it reaches the value of antilog (–2.273), which with $a_{\mathrm{MgO}} = 1$ makes $a_{\mathrm{MgAl_2O_4}} = 1$.
- ✓ Thus, the four-phase equilibrium melt–MgO–$MgAl_2O_4$–gas occurs at the point *b*.

13.7 Graphical Representation Of Phase Equilibria

- ✓ The three-phase equilibrium involving the Mg–Al melt, $MgAl_2O_4$, and the gas phase, determined by Equation (iiia), occurs along the line *bd* .
- ✓ In moving along the line from *b* to *d*, the dilution of Mg in the melt causes a_{MgO} to decrease from unity at *b* and $a_{\mathrm{Al_2O_3}}$ to increase, and at the point *d*, $a_{\mathrm{Al_2O_3}} = 1$ and $\log a_{\mathrm{MgO}} = -2.273$.
- ✓ Thus, *d* represents the four-phase equilibrium melt–Al_2O_3–$MgAl_2O_4$–gas.
- ✓ At activities of Mg less than that at *d*, the activity of MgO is less than antilog (–2.273), and thus, the spinel $MgAl_2O_4$ is not stable. The line *ed* represents the equilibrium involving a melt, Al_2O_3, and a gas phase, given by Equation (ia).
- ✓ However, as $a_{\mathrm{Mg}} = 5.75\times10^{-5}$ at the point *d*, a_{Al} is virtually unity, and thus, the line *ed* is virtually horizontal at $\log p_{\mathrm{O_2}}$ (obtained from Equation [ia]) = –42.93.

13.7 Graphical Representation Of Phase Equilibria

- ✓ The lines *df* and *bc* represent, respectively, the three-phase equilibria Al_2O_3–$MgAl_2O_4$–gas and $MgAl_2O_4$–MgO–gas.
- ✓ The lines in Figure 13.18a identify the fields of stability of Al–Mg liquid alloys, MgO, $MgAl_2O_4$, and Al_2O_3 at 1073 K.
- ✓ The activity–composition relationships given by Equations (v) and (vi) allow the phase stability diagram presented as Figure 13.18a to be converted to the phase diagram shown in Figure 13.18b, in which $\log p_{O_2}$ and $\log n_{Mg}/(n_{Mg}+n_{Al})$ (where n_{Mg} and n_{Al} are, respectively, the numbers of moles of Mg and Al in the system) are used as coordinates.
- ✓ The line *ab* gives the compositions of the metallic melts saturated with MgO, the line *bd* gives the compositions of the melts saturated with $MgAl_2O_4$, and the line *ed* gives the compositions of the melts saturated with Al_2O_3.

13.7 Graphical Representation Of Phase Equilibria

- ✓ Consider the sequence of oxidation of Al–Mg alloys of X_{Mg} = 0.333, 0.2, and 0.01 at 1073 K.
- ✓ With X_{Mg} = 0.333 and $\log p_{O_2} = -48$, the system exists at the state *a* in Figure 13.19. When the oxygen pressure is increased to $\log p_{O_2} = -45.6$, the system exists at *b*, in which state the alloy is in equilibrium with MgO.
- ✓ Further increase in the oxygen pressure causes the precipitation of MgO from the melt, which decreases the mole fraction of X Mg in the melt and causes its composition to move along the MgO saturation line from *b* toward *c*.
- ✓ At $\log p_{O_2} = -44.3$, the melt of composition *c* is saturated with MgO, and further increase in the oxygen pressure causes all of the melt of composition *c* to react with all of the MgO at d to form the spinel $MgAl_2O_4$.

13.7 Graphical Representation Of Phase Equilibria

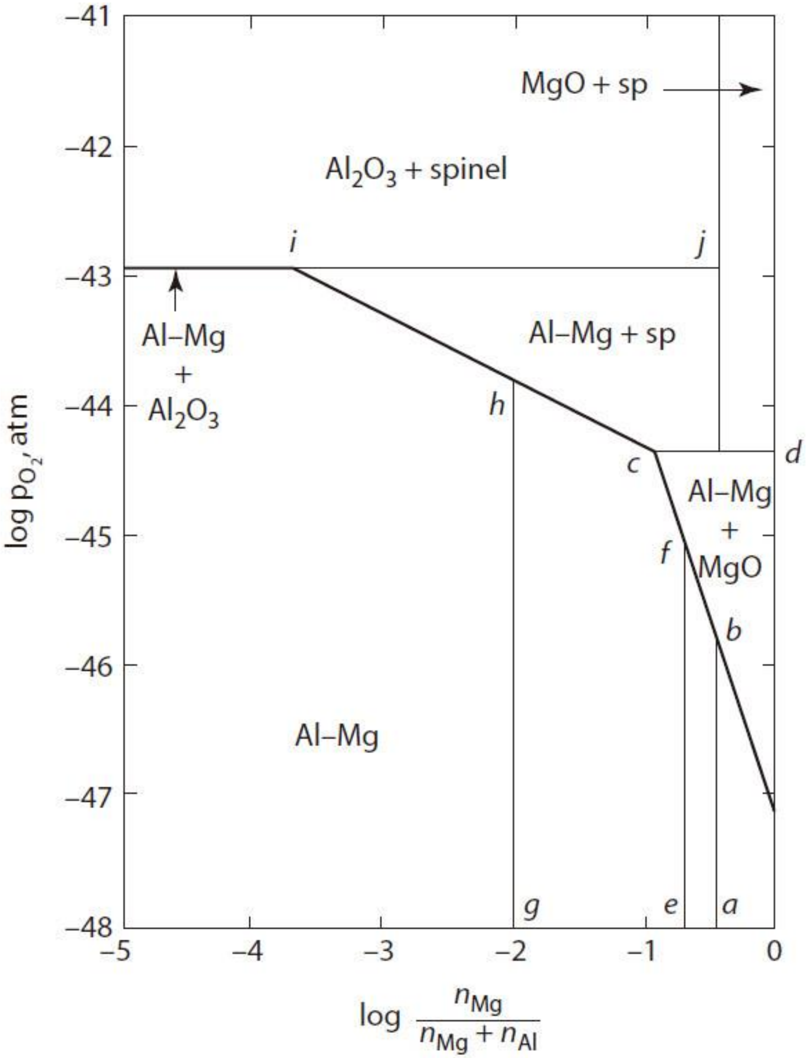

Figure 13.19 The influence of composition and oxygen pressure on the equilibrium states of existence in the system Mg–Al–O.

13.7 Graphical Representation Of Phase Equilibria

- ✓ A melt of X_{Mg} = 0.2 equilibrated with oxygen at $\log p_{O_2} = -48$ exists at the state *e* in Figure 13.19, and increasing the oxygen pressure to $\log p_{O_2} = -45.1$ brings the alloy into equilibrium with MgO at the state *f*.
- ✓ Further increase in the oxygen pressure to $\log p_{O_2} = -44.3$ causes MgO to precipitate and moves the composition of the melt along the MgO saturation line from *f* to *c*.
- ✓ Further increase in the oxygen pressure causes all of the MgO to react with some of the melt to produce $MgAl_2O_4$ and a melt saturated with $MgAl_2O_4$.
- ✓ With further oxidation, $MgAl_2O_4$ is precipitated from the melt, and the composition of the melt is moved along the $MgAl_2O_4$ saturation line from *c* toward *i*.

13.7 Graphical Representation Of Phase Equilibria

- ✓ Although the precipitation of the spinel from the melt removes Al and Mg atoms from the melt in the ratio Al/Mg = 2, the values of X_{Mg} in the melts are low enough that the removal of Al atoms makes a negligible change in X_{Al} but causes a significant decrease in X_{Mg}.
- ✓ At $\log p_{O_2} = -43.8$, the melt at the state *i* is doubly saturated with Al_2O_3 and $MgAl_2O_4$, and further increase in the oxygen pressure causes the disappearance of the melt.
- ✓ Increasing the oxygen pressure exerted on a melt of X_{Mg} = 0.01 causes saturation of the melt with $MgAl_2O_4$ at $\log p_{O_2} = -43.8$ (the state *h*), and further increase in the oxygen pressure causes oxidation to proceed as described previously.

13.7.2 Phase Equilibria in the System Al–C–O–N Saturated with Carbon

- ✓ We now consider phase stability in the system Al–C–O–N, saturated with carbon at 2000 K, and will identify the conditions under which AlN can be contained in a graphite crucible at 2000 K without the formation of Al_4C_3.
- ✓ The solid phases which occur in the quaternary system are AlN, Al_4C_3, Al_4O_4C, and Al_2O_3, and since the system is saturated with graphite, the minimum number of phases which can coexist in equilibrium with one another is three (graphite, a gas phase, and a second condensed phase).
- ✓ The number of degrees of freedom available to this three-phase equilibrium is

$$F = C + 2 - \Phi = 4 + 2 - 3 = 3$$

which can be selected as T, p_{O_2}, and p_{N_2}.

13.7 Graphical Representation Of Phase Equilibria

- ✓ An isothermal phase stability diagram can thus be constructed using $\log p_{N_2}$, and $\log p_{O_2}$ as coordinates.
- ✓ For

$$4Al_{(l)} + 3C_{(s)} = Al_4C_{3(s)} \qquad \text{(i)}$$

$$\Delta G^\circ_{(i),2000\ K} = -74,060\ J$$

and thus, an equilibrium involving pure liquid Al, graphite, and solid Al_4C_3 does not exist at 2000 K.

- ✓ The activity of Al in carbon-saturated Al_4C_3 is obtained from

$$K_{(i),2000\ K} = \exp\left(\frac{74,060}{8.3144 \times 2000}\right) = 85.95 = \frac{a_{Al_4C_3}}{a_{Al}^4 a_C^3} = \frac{1}{a_{Al}^4}$$

which gives $a_{Al} = 0.327$, relative to pure liquid Al.

13.7 Graphical Representation Of Phase Equilibria

- For

$$4\text{AlN}_{(s)} + 3\text{C}_{(s)} = \text{Al}_4\text{C}_{3(s)} + 2\text{N}_{2(g)} \qquad \text{(ii)}$$

$$\Delta G^\circ_{\text{(ii)},2000\text{ K}} = 917,900\text{ J} = -2\times 8.3144\times 2000\times 2.303\log p_{\text{N}_2}$$

which gives $\log p_{\text{N}_2} = -11.98$ for the equilibrium involving solid graphite, solid AlN, solid Al_4C_3, and a gas phase.

- The line representing this equilibrium is drawn as *ab* in Figure 13.20a.
- Graphite and AlN are stable relative to graphite and Al_4C_3 in states above the line, and graphite and Al_4C_3 are stable relative to graphite and AlN in states below the line.

13.7 Graphical Representation Of Phase Equilibria

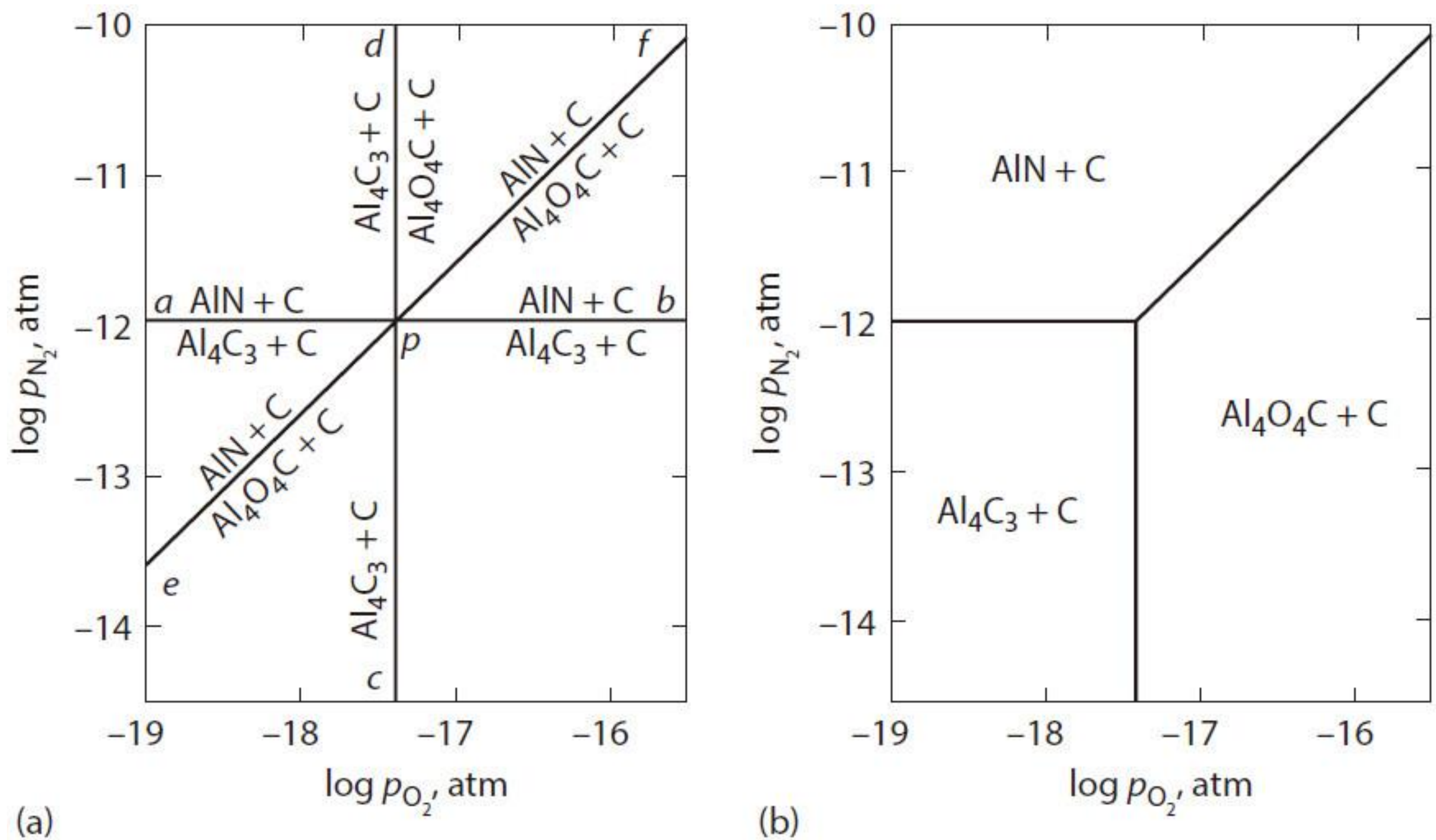

Figure 13.20 Construction of the phase stability diagram for the system Al–C–O–N saturated with carbon at 2000 K.

13.7 Graphical Representation Of Phase Equilibria

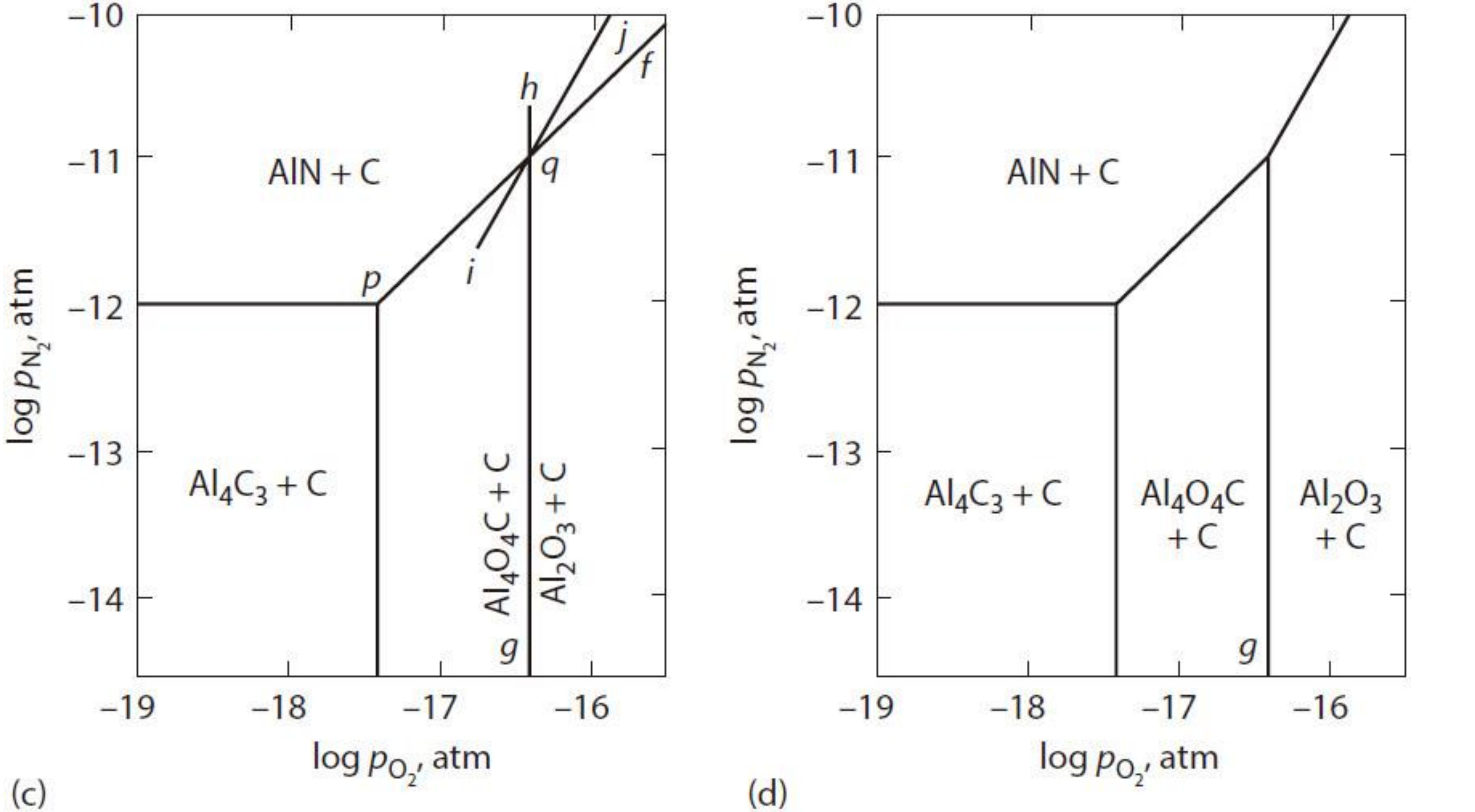

Figure 13.20 Construction of the phase stability diagram for the system Al–C–O–N saturated with carbon at 2000 K.

13.7 Graphical Representation Of Phase Equilibria

- ✓ For

$$Al_4O_4C_{(s)} + 2C_{(s)} = Al_4C_{3(s)} + 2O_{2(g)} \qquad \text{(iii)}$$

$$\Delta G^\circ_{\text{(iii)},2000\text{ K}} = 1,333,000 \text{ J} = -2 \times 8.3144 \times 2000 \times 2.303 \log p_{O_2}$$

which gives $\log p_{O_2} = -17.40$ for the equilibrium involving solid graphite, solid Al_4C_3, solid Al_4O_4C, and a gas phase.

- ✓ This is drawn as line *cd* in Figure 13.20a.

- ✓ To the left of this line, graphite and Al_4C_3 are stable relative to graphite and Al_4O_4C, and to the right of the line, the reverse is the case.

13.7 Graphical Representation Of Phase Equilibria

- ✓ For

$$4\mathrm{Al}_{(l)} + 2\mathrm{O}_{2(g)} + \mathrm{C}_{(s)} = \mathrm{Al}_4\mathrm{O}_4\mathrm{C}_{(s)} \qquad \text{(iv)}$$

$$\Delta G^{\circ}_{\text{(iv)},2000\ \text{K}} = -1,407,000\ \text{J}$$

and thus, for carbon-saturated Al_4O_4C,

$$K_{\text{(iv)},2000\ \text{K}} = \exp\left(\frac{1,407,000}{8.3144\times 2000}\right) = 5.58\times 10^{36} = \frac{1}{a_{\mathrm{Al}}^4 p_{\mathrm{O}_2}}$$

which, with $\log p_{\mathrm{O}_2} = -17.40$, gives a_{Al} = 0.327.

- ✓ Thus, the activity of Al in C-saturated Al_4O_4C has the same value as that in C-saturated AlN.
- ✓ The point of intersection of lines *ab* and *cd* is the state in which the five-phase equilibrium involving four solid phases and a gas phase occurs.

13.7 Graphical Representation Of Phase Equilibria

- For

$$\mathrm{Al_4O_4C}_{(s)} + 2\mathrm{N}_{2(g)} = \mathrm{C}_{(s)} + 2\mathrm{O}_{2(g)} + 4\mathrm{AlN}_{(s)} \qquad \text{(v)}$$

$$\Delta G^{\circ}_{(\mathrm{v}),2000\ \mathrm{K}} = 415,000\ \mathrm{J} = -2\times 8.3144\times 2000\times 2.303\log\frac{p_{\mathrm{O_2}}}{p_{\mathrm{N_2}}}$$

which gives

$$\log p_{\mathrm{N_2}} = \log p_{\mathrm{O_2}} + 5.42$$

for the equilibrium among graphite, solid AlN, solid Al_4O_4C, and a gas phase.

- This is drawn as line *ef* in Figure 13.20a.
- Carbon-saturated AlN is stable relative to carbon- saturated Al_4O_4C above the line, with the reverse being the case below the line.

13.7 Graphical Representation Of Phase Equilibria

- ✓ Inspection of Figure 13.20a shows the following:

1. In the area below *ap* and to the left of *pc*, Al_4C_3 is stable with respect to AlN and Al_4O_4C.
2. In the area above *ap* and above *pf*, AlN is stable with respect to Al_4C_3 and Al_4O_4C.
3. In the area below *pf* and to the right of *pc*, Al_4O_4C is stable with respect to AlN and Al_4C_3.

- ✓ Thus, without consideration of the stability of Al_2O_3, the fields of stability of AlN, Al_4C_3, and Al_4O_4C are as shown in Figure 13.20b.

13.7 Graphical Representation Of Phase Equilibria

- For

$$2Al_2O_{3(s)} + C_{(s)} = O_{2(g)} + Al_4O_4C_{(s)} \quad \text{(vi)}$$

$$\Delta G^\circ_{(vi),2000\ K} = 630,800\ J = -8.3144 \times 2000 \times 2.303 \log p_{O_2}$$

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which gives $\log p_{O_2} = -16.47$ for the equilibrium among graphite Al_4O_4C, Al_2O_3, and a gas phase. .

- This is drawn as the line *gh* in Figure 13.20c, and its point of intersection with *pf* is the state of the five-phase equilibrium among graphite, AlN, Al_4O_4C, Al_2O_3, and a gas phase

13.7 Graphical Representation Of Phase Equilibria

- ✓ For

$$Al_2O_{3(s)} + N_{2(g)} = 2AlN_{(s)} + 1.5O_{2(g)} \qquad \text{(vii)}$$

$$\Delta G^{\circ}_{\text{(vii)},2000\text{ K}} = 522,900 \text{ J} = -8.3144 \times 2000 \times 2.303 \log \frac{p_{O_2}^{1.5}}{p_{N_2}}$$

which gives

$$\log p_{N_2} = 1.5 \log p_{O_2} + 13.65$$

for the equilibrium given by Equation (vii).

- ✓ This line, which is independent of the activity of C in the system, is drawn as *ij* in Figure 13.20c.
- ✓ Since the line *pq* represents a stable equilibrium, the other two lines radiating from the point *q* which represent stable equilibria are *qg* and *gj*, and the full phase stability diagram is as shown in Figure 13.20d.
- ✓ The diagram shows that AlN can be heated in a graphite crucible without forming Al_4C_3 if the pressures of oxygen and nitrogen are such that the thermodynamic state lies in the field of stability of AlN.

Outlines

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ Figure 13.21 shows the variation, with composition at a temperature *T*, of the molar Gibbs free energy of mixing for the metal M–oxygen system in which measurable solubility of oxygen in metallic M occurs and the oxides MO and M_3O_4 have variable compositions.
- ✓ Compounds with nonstoichiometric compositions are sometimes called *berthollides*, after Claude Louis Berthollet (1748–1822).
- ✓ Starting with pure M, increasing the pressure of oxygen causes the molar Gibbs free energy of mixing to move from *f* along the line *fi* until, at $p_{O_2} = p_{O_2(M/MO)}$, the metal is saturated with oxygen and the metal-saturated MO phase of composition M_bO appears.

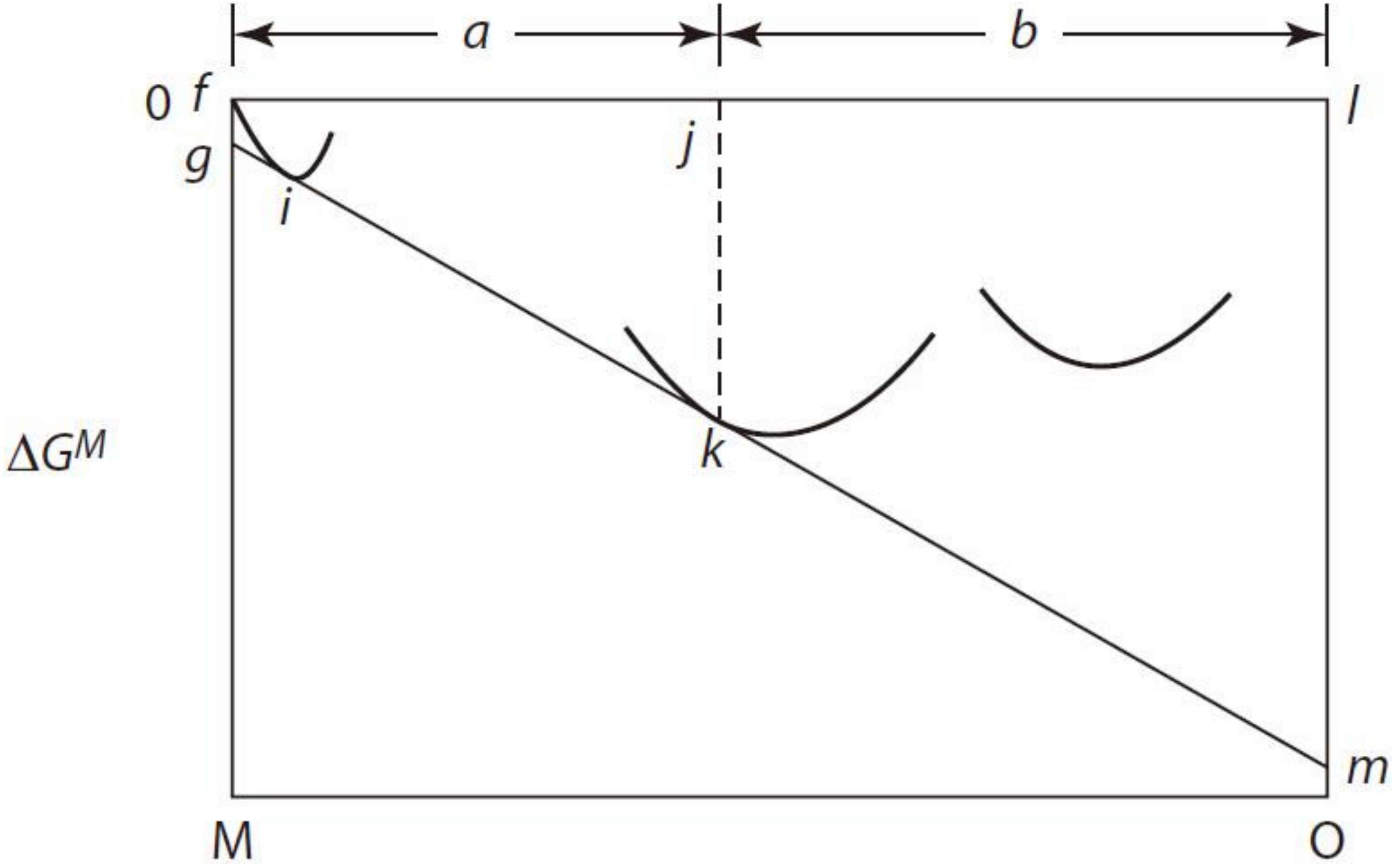

Figure 13.21 The Gibbs free energies of mixing of the system M–O which forms oxide phases of variable composition and which shows a significant solubility of oxygen in metallic M.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ If the pure metal M and oxygen gas at 1 atm pressure at the temperature *T* are chosen as the standard states, then

$$\Delta G_{(\text{metal/saturated MO})} = jk = RT\left(b\ln a_{\text{M}} + a\ln p_{\text{O}_2}^{1/2}\right) = RT\ln a_{\text{M}}^{b} p_{\text{O}_2}^{1/2a}$$

- ✓ From Figure 13.21, $fg = RT\ln a_{\text{M}}$ and $lm = RT\ln p_{\text{O}_2\,(\text{M/MO})}$.
- ✓ We can also write for the reaction

$$b\text{M}_{(s)} + \frac{1}{2}a\text{O}_{2(g)} = \text{M}_b\text{O}_{a(s)}$$

$$\Delta G^{\circ} = jk = RT\ln\left(\frac{a_{\text{M}}^{b} p_{\text{O}_2}^{1/2a}}{a_{\text{M}_b\text{O}_a}}\right)$$

which, if the metal-saturated oxide is chosen as the standard state, is identical with the preceding expression.

- ✓ It is convenient to write the reaction of the oxidation such that an integer number of gram-atoms of oxygen are consumed.

- ✓ For example, for the consumption of 1 gram-atom of oxygen,

$$y\mathrm{M}+\frac{1}{2}\mathrm{O}_2=\mathrm{M}_y\mathrm{O},\qquad \text{where } y=\frac{b}{a}$$

$$\Delta G^{\circ}=RT\ln\left(\frac{a_{\mathrm{M}}^{y}\,p_{\mathrm{O}_2}^{1/2}}{a_{\mathrm{M}_y\mathrm{O}}}\right)$$

or, for the consumption of 1 gram-mole of oxygen (2 gram-atoms),

$$\Delta G^{\circ}=RT\ln\left(\frac{a_{\mathrm{M}}^{2y}\,p_{\mathrm{O}_2}}{a_{\mathrm{M}_y\mathrm{O}}^{2}}\right)$$

- ✓ If the solubility of oxygen in the metal is virtually zero, then *fg* in Figure 13.21 shrinks to a point, and Figure 13.21 is redrawn as Figure 13.22.

13.8 The Formation Of Oxide Phases Of Variable Composition

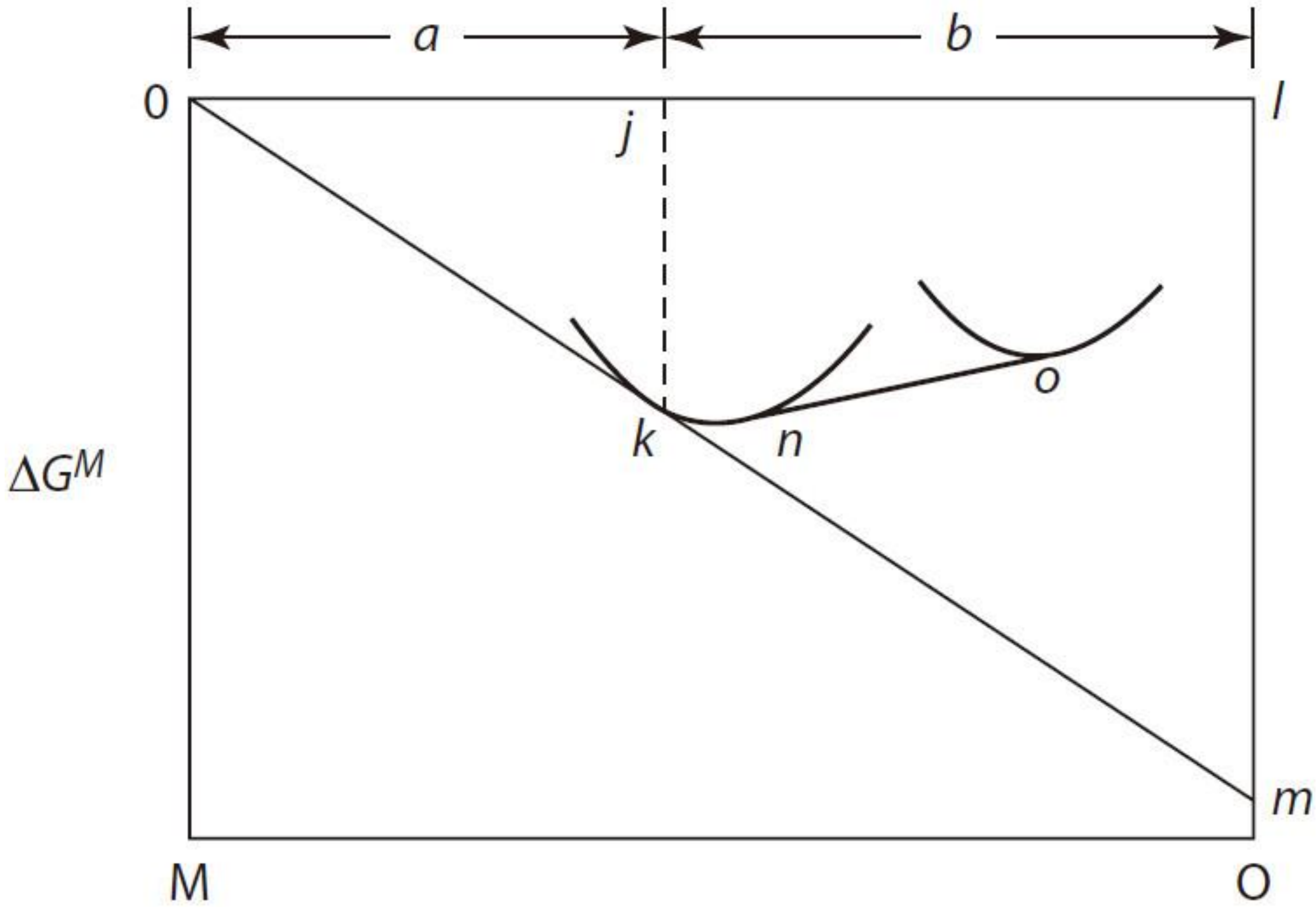

Figure 13.22 The Gibbs free energies of mixing of the system M–O which forms oxide phases of variable composition and which shows a negligible solubility of oxygen in metallic M.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ In this case, choosing pure M, oxygen gas at 1 atm pressure and the temperature *T*, and the oxide of composition M_yO as the standard states gives, for the oxidation,

$$y\mathrm{M} + \frac{1}{2}\mathrm{O}_2 = \mathrm{M}_y\mathrm{O}$$

$$\Delta G^{\circ} = RT \ln p^{1/2}_{\mathrm{O}_2(\mathrm{M/MO})}$$

where $p_{\mathrm{O}_2(\mathrm{M/MO})}$ is the pressure of oxygen required for equilibrium between the metal M and the metal-saturated oxide MO, given by $lm = RT \ln p_{\mathrm{O}_2(\mathrm{M/MO})}$.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ If the oxygen pressure is increased to a value greater than $p_{O_2(M/MO)}$, the metal phase disappears, the oxygen content of the MO phase increases, the molar Gibbs free energy of mixing of the system moves along the line *kn*, and the activities of M and MO vary accordingly.
- ✓ In a classic investigation, Darken and Gurry determined the phase relationships occurring in the system Fe–O by varying the oxygen pressure and temperature and observing the consequential changes in phase and phase composition.
- ✓ Their diagram, drawn for the components FeO and Fe_2O_3, is shown in Figure 13.23.
- ✓ Consider the wustite ("FeO")* phase field which, at 1100° C, extends from the composition *m* to the composition *n*.

*Quotation marks are used for wustite since the its composition is only approximately equiatomic.

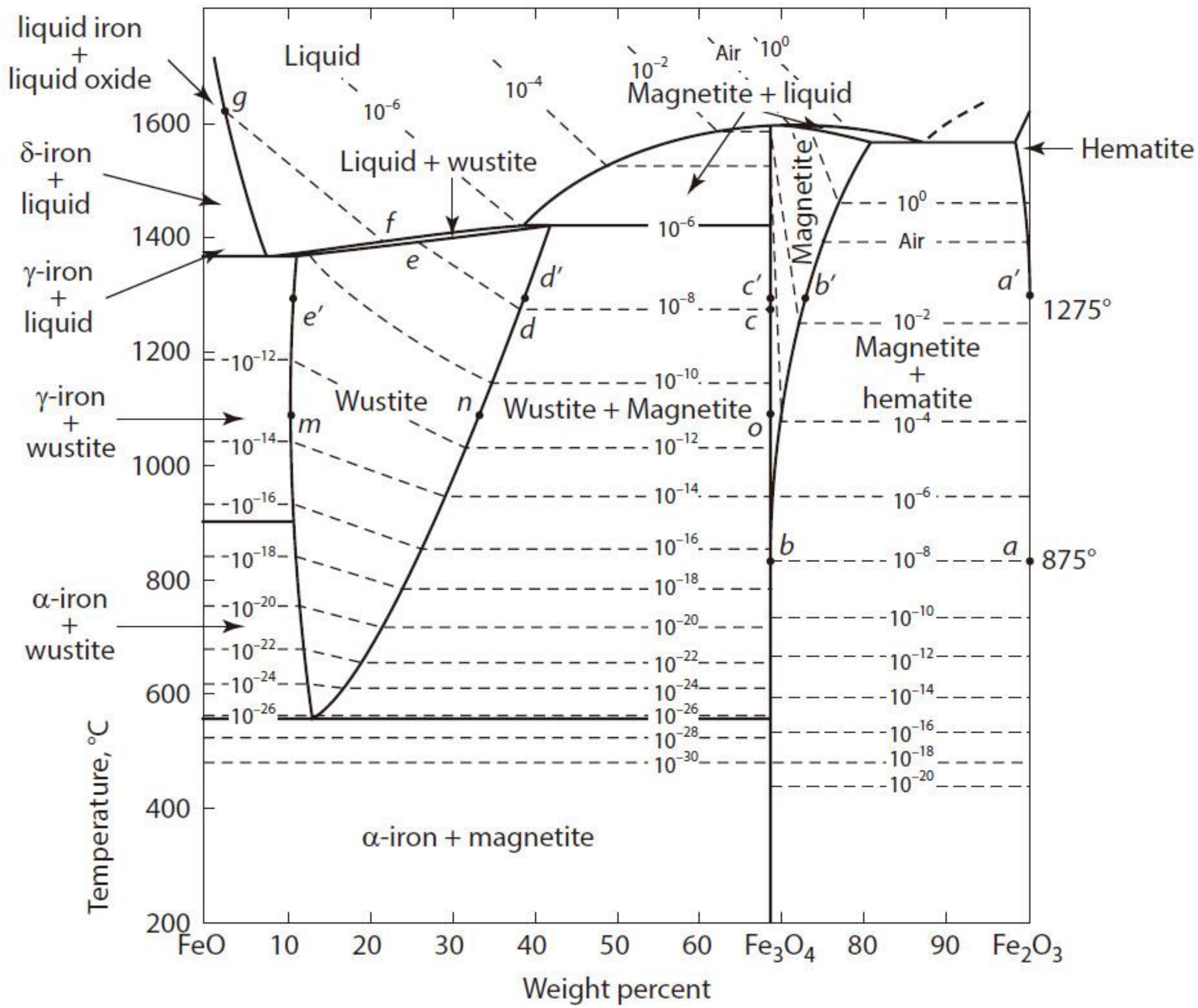

Figure 13.23 The phase diagram for the system FeO–Fe_2O_3 showing the positions of the oxygen (atm) isobars.

- ✓ The variation of a_{Fe} in the wustite phase can be calculated from the experimentally determined variation of the composition of wustite with oxygen pressure using the Gibbs–Duhem equation:

$$X_{\mathrm{Fe}} d \ln a_{\mathrm{Fe}} + X_{\mathrm{O}} d \ln a_{\mathrm{O}} = 0$$

- ✓ That is,

$$\log a_{\mathrm{Fe}} = -\int \frac{X_{\mathrm{O}}}{X_{\mathrm{Fe}}} d \log a_{\mathrm{O}} = -\int \frac{X_{\mathrm{O}}}{X_{\mathrm{Fe}}} d \log p_{\mathrm{O}_2}^{1/2}$$

where the upper limit of integration is the oxygen pressure in equilibrium with the wustite composition of interest, and the lower limit is $p_{\mathrm{O}_2\,(\mathrm{Fe/"FeO"})}$ (the oxygen pressure at which wustite of composition *m* is in equilibrium with oxygen-saturated metallic iron), at which composition $a_{\mathrm{Fe}} = 1$.

- Having thus determined the variations of a_{Fe} with composition, the corresponding variations of $a_{\text{"FeO"}}$ are determined as follows.
- If the standard state for oxygen gas is selected as being $p_{O_2(\text{Fe/"FeO"})}$ at the temperature of interest, then for

$$y\text{Fe}_{(s)} + \frac{1}{2}\text{O}_{2(g,\ \text{at } p_{O_2(\text{Fe/"FeO"})})} = \text{Fe}_y\text{O}_{(s)}$$

as the standard states are in equilibrium with one another, $\Delta G^\circ = 0$, and thus, K = 1,

$$a_{\text{"FeO"}} = a_{\text{Fe}} a_{\text{O}} \qquad \text{where } a_{\text{O}} = \left(\frac{p_{O_2}}{p_{O_2(\text{Fe/"FeO"})}} \right)^{1/2}$$

or

$$\log a_{\text{"FeO"}} = \log a_{\text{Fe}} + \log a_{\text{O}} = -\int \frac{X_{\text{O}}}{X_{\text{Fe}}} d\log a_{\text{O}} + \int d\ \text{og}\ a_{\text{O}} = -\int \left(\frac{X_{\text{O}}}{X_{\text{Fe}}} - 1 \right) d\log p_{O_2}^{1/2}$$

in which the integration limits are the same as before.

- The variations of a_{Fe} , $a_{\text{"FeO"}}$, and a_{O} across the wustite field at 1100°, 1200°, and 1300 ℃ are shown in Figure 13.24, and from these variations, the molar Gibbs free energy curve for wustite, *kn* in Figure 13.22, can be determined.
- For a fixed composition, the partial molar heats of solution of metal and oxygen in the wustite can be obtained from the Gibbs–Helmholtz relationship as

$$\Delta \overline{H}_{\text{O}}^{M} = R\frac{\partial \ln p_{\text{O}_2}^{1/2}}{\partial\left(1/T\right)}$$

and

$$\Delta \overline{H}_{\text{Fe}}^{M} = R\frac{\partial \ln a_{\text{Fe}}}{\partial\left(1/T\right)}$$

- The variations of $\Delta \overline{H}_{\text{O}}^{M}$ and $\Delta \overline{H}_{\text{Fe}}^{M}$ with composition are shown in Figure 13.25.

13.8 The Formation Of Oxide Phases Of Variable Composition

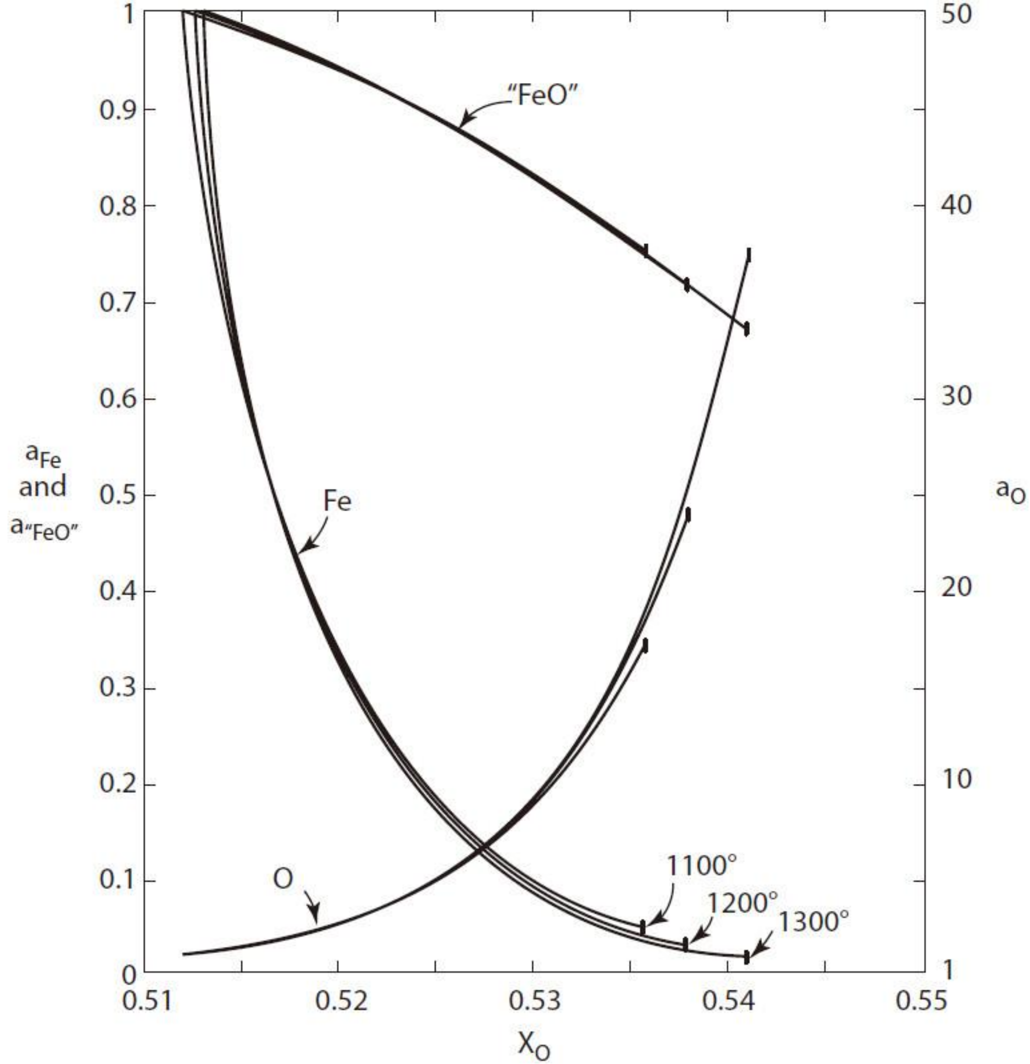

Figure 13.24 The activities of iron, oxygen, and iron-saturated wustite in the wustite phase field at several temperatures.

13.8 The Formation Of Oxide Phases Of Variable Composition

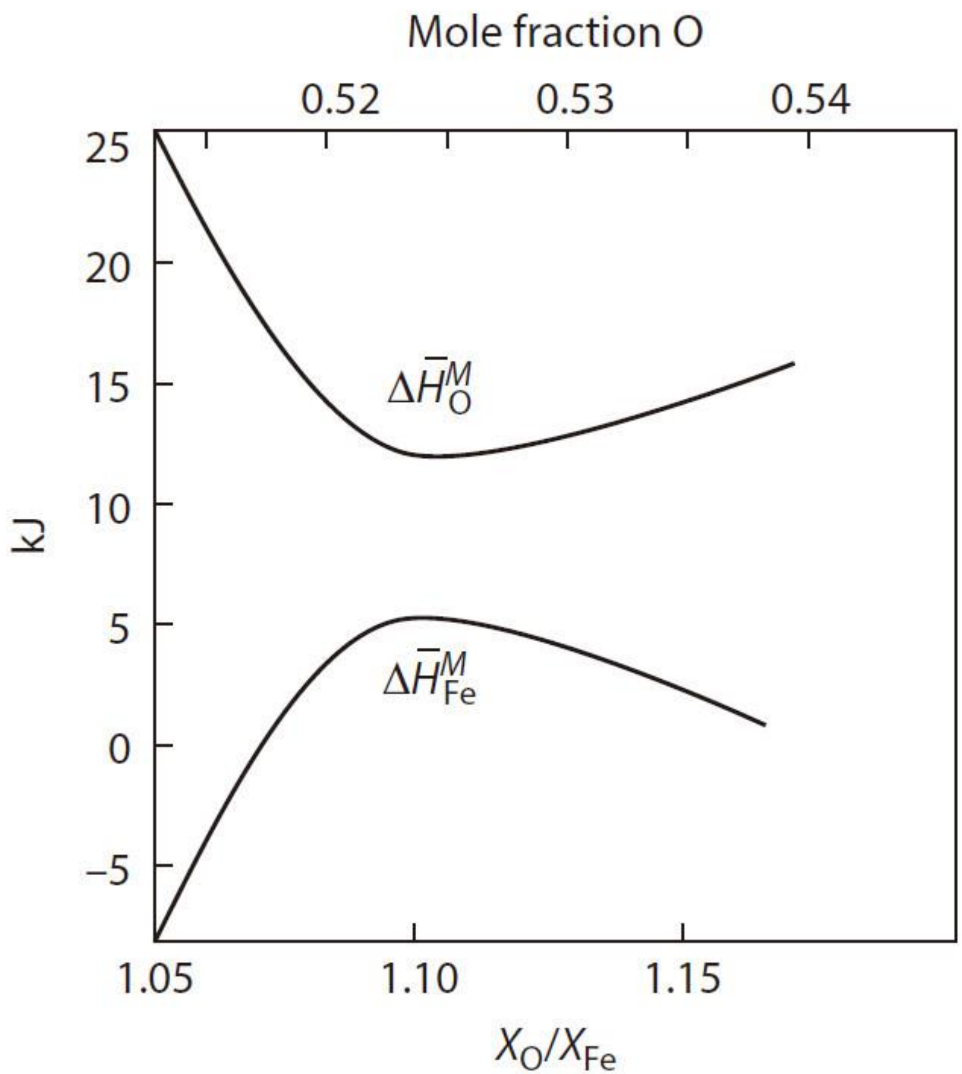

Figure 13.25 The partial molal heats of solution of iron and oxygen in wustite.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ In Figure 13.24, at the composition of wustite X_O = 0.5125, the activity of Fe is independent of temperature, and thus, as shown in Figure 13.25, the partial molar heat of mixing of Fe at this composition is zero.
- ✓ At the temperature *T*, the limit of increase of p_{O_2} above homogeneous stable wustite is $p_{O_2(\text{"FeO"}/Fe_3O_4)}$, the oxygen pressure at which wustite of composition *n* is in equilibrium with magnetite (Fe_3O_4) of composition *o* (Figure 13.23), and, if these compositions are chosen as the standard states, then for

$$3\text{"FeO"}_{(\text{saturated with oxygen})} + \frac{1}{2}O_{2(g)} = Fe_3O_{4(\text{saturated with Fe})}$$

$$\Delta G^\circ = RT \ln p^{1/2}_{O_2(\text{"FeO"}/Fe_3O_4)}$$

- ✓ Figure 13.23 shows that the composition of wustite in equilibrium with magnetite varies significantly with temperature.

- Thus, the heat of formation of magnetite from wustite cannot be calculated by application of the Gibbs–Helmholtz equation to the variation of $p_{O_2(\text{"FeO"}/Fe_3O_4)}$ with temperature (the Gibbs– Helmholtz partial differential is for constant total pressure and constant composition).
- However, since the composition of magnetite in equilibrium with wustite is independent of temperature, the change in enthalpy for the reaction

$$3Fe_{(s)} + 2O_{2(g)} = Fe_3O_{4(\text{saturated with Fe})}$$

can be obtained using the Gibbs–Helmholtz relationship; that is,

$$K = \frac{1}{a_{Fe}^3 p_{O_2}^2}$$

and thus,

$$\Delta H^\circ = -R\frac{d\ln K}{d(1/T)} = R\left[3\frac{d\ln a_{Fe}}{d(1/T)} + 2\frac{d\ln p_{O_2}}{d(1/T)}\right]$$

- In this expression, a_{Fe} and p_{O_2} are the respective values for equilibrium between wustite and magnetite obtained from the data in Figure 13.24.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ Below 550 ℃, homogeneous wustite is metastable with respect to iron and magnetite.
- ✓ This situation corresponds to Figure 13.11b in that, below 550 ℃(T_2 in Figure 13.11; see also Figure 13.8), the common tangent drawn from pure Fe (B in Figure 13.11) to the curve for the Gibbs free energy of magnetite lies below the curve for wustite.
- ✓ At 550 ℃, this common tangent becomes a “ triple” common tangent and the two-component, four-phase equilibrium is invariant (Figure 13.11a).
- ✓ The phase diagram at 1 atm total pressure shown in Figure 13.23 has, superimposed on it, oxygen isobars which trace the loci of variation of equilibrium composition with temperature under a fixed oxygen pressure in the system.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ For example, consider a small quantity of hematite (Fe_2O_3) at room temperature held in a gas reservoir of $p_{O_2} = 10^{-8}$ atm, the volume of which is sufficiently large that any oxygen gas produced by the reduction of the oxide has an insignificant effect on the pressure of oxygen in the gas reservoir.
- ✓ Let the oxide be heated slowly enough that equilibrium with the gas phase is maintained.
- ✓ From Figure 13.23, it is seen that the oxide remains as homogeneous hematite until 875 ℃ is reached, at which temperature 10^{-8} atm is the invariant partial pressure of oxygen required for the equilibrium

$$2Fe_3O_4 + \frac{1}{2}O_2 = 3Fe_2O_3$$

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ At 875 ℃, magnetite of composition *b* is in equilibrium with hematite of composition *a*, and any increase in temperature upsets the equilibrium toward the magnetite side, with the consequent disappearance of the hematite phase.
- ✓ Further increase in temperature moves the composition of the oxide along the 10^{-8} atm isobar in the magnetite phase field until 1275 ℃ is reached, at which temperature 10^{-8} atm is the invariant partial pressure of oxygen required for the equilibrium

$$3\text{"FeO"} + \frac{1}{2}O_2 = Fe_3O_4$$

- ✓ At 1275 ℃ wustite of composition *d* is in equilibrium with magnetite of composition *c*.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ Further increase in temperature causes the disappearance of the magnetite phase, and the composition of the solid homogeneous wustite moves along the 10^{-8} atm oxygen isobar until the solidus temperature of 1400 ℃ is reached, in which state solid wustite of composition *e* melts to form a liquid oxide of composition *f* at $p_{O_2} = 10^{-8}$ atm.
- ✓ Continued increase in temperature moves the composition of the liquid oxide along the 10^{-8} atm isobar to saturation with iron at the temperature 1635 ℃, where the liquid oxide has the composition *g*, and oxygen-saturated liquid iron appears.
- ✓ In this state, the equilibrium

$$\mathrm{Fe}_{(l)} + \frac{1}{2}\mathrm{O}_{2(g)} = \text{"FeO"}_{(l)}$$

is established.

- ✓ An increase in temperature beyond 1635 ℃ causes the disappearance of the liquid oxide phase and a decrease in the dissolved oxygen content of the liquid iron.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ Similarly, isothermal reduction of hematite is achieved by decreasing the partial pressure of oxygen in the system.

- ✓ For example, from Figure 13.23, at 1300 ℃, hematite is the stable phase until the partial pressure of oxygen has been decreased to 1.34×10^{-2} atm, in which state magnetite of composition *b′* is in equilibrium with hematite of composition *a′*.

- ✓ Magnetite is then stable until the partial pressure of oxygen has been decreased to 2.15×10^{-8} atm, where wustite of composition *d′* is in equilibrium with magnetite of composition *c′*.

- ✓ Wustite is then stable until the partial pressure of oxygen has been decreased to 1.95×10^{-11} atm, where solid iron appears in equilibrium with wustite of composition *e′*.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ Further decrease in the pressure of oxygen causes the disappearance of the oxide phase.
- ✓ Figure 13.26 shows the phase relationships in a plot of $\log p_{O_2}$ versus temperature *T*, and the paths *a*–*g* and *a'*–*e'* correspond to those in Figure 13.23.
- ✓ In that Figure 13.26 does not contain the compositions of the coexisting oxide phases, it is less useful than the normal composition–temperature phase diagram containing oxygen isobars.
- ✓ Figure 13.27 shows the phase equilibria on a plot of 1/*T* versus $\log p_{O_2}$.
- ✓ This plot can be converted to an Ellingham diagram, as shown in Figure 13.28. In this diagram, the paths *a*–*g* and *a'*–*e'* correspond to those in Figure 13.23.

13.8 The Formation Of Oxide Phases Of Variable Composition

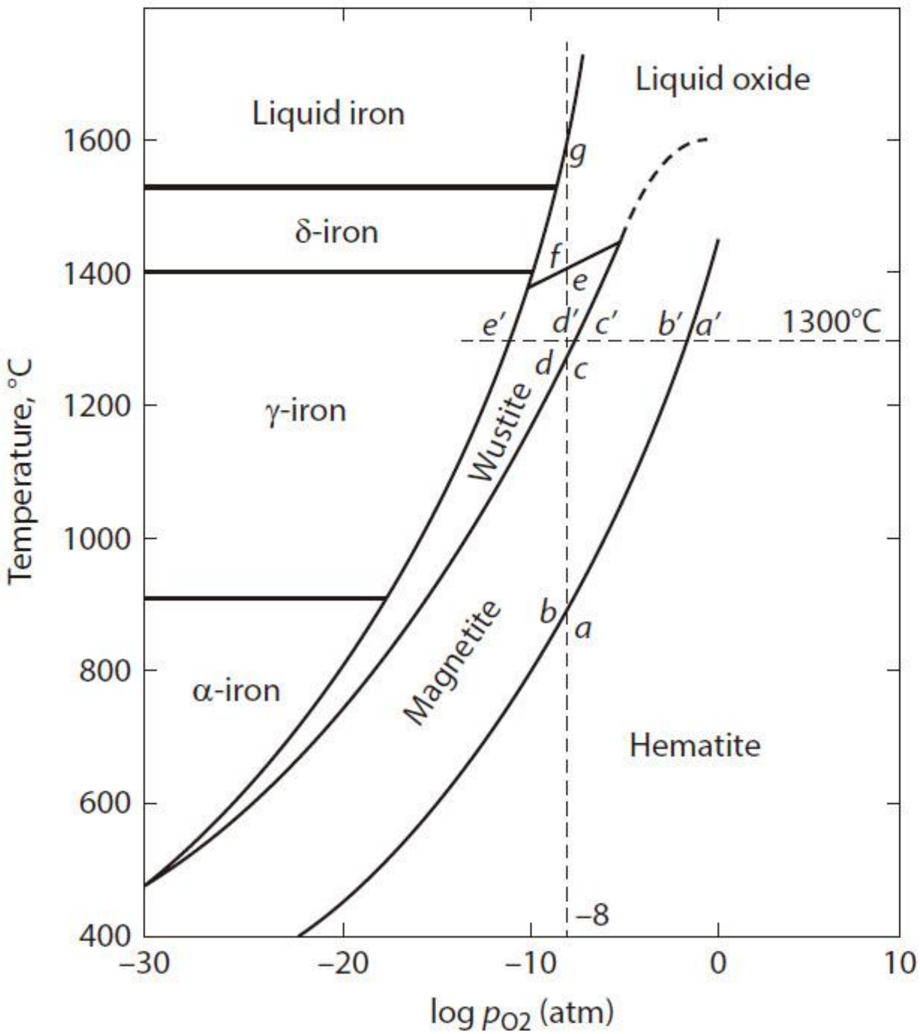

Figure 13.26 Phase stability in the system $Fe{-}Fe_2O_3$ as a function of temperature and $\log p_{O_2}$.

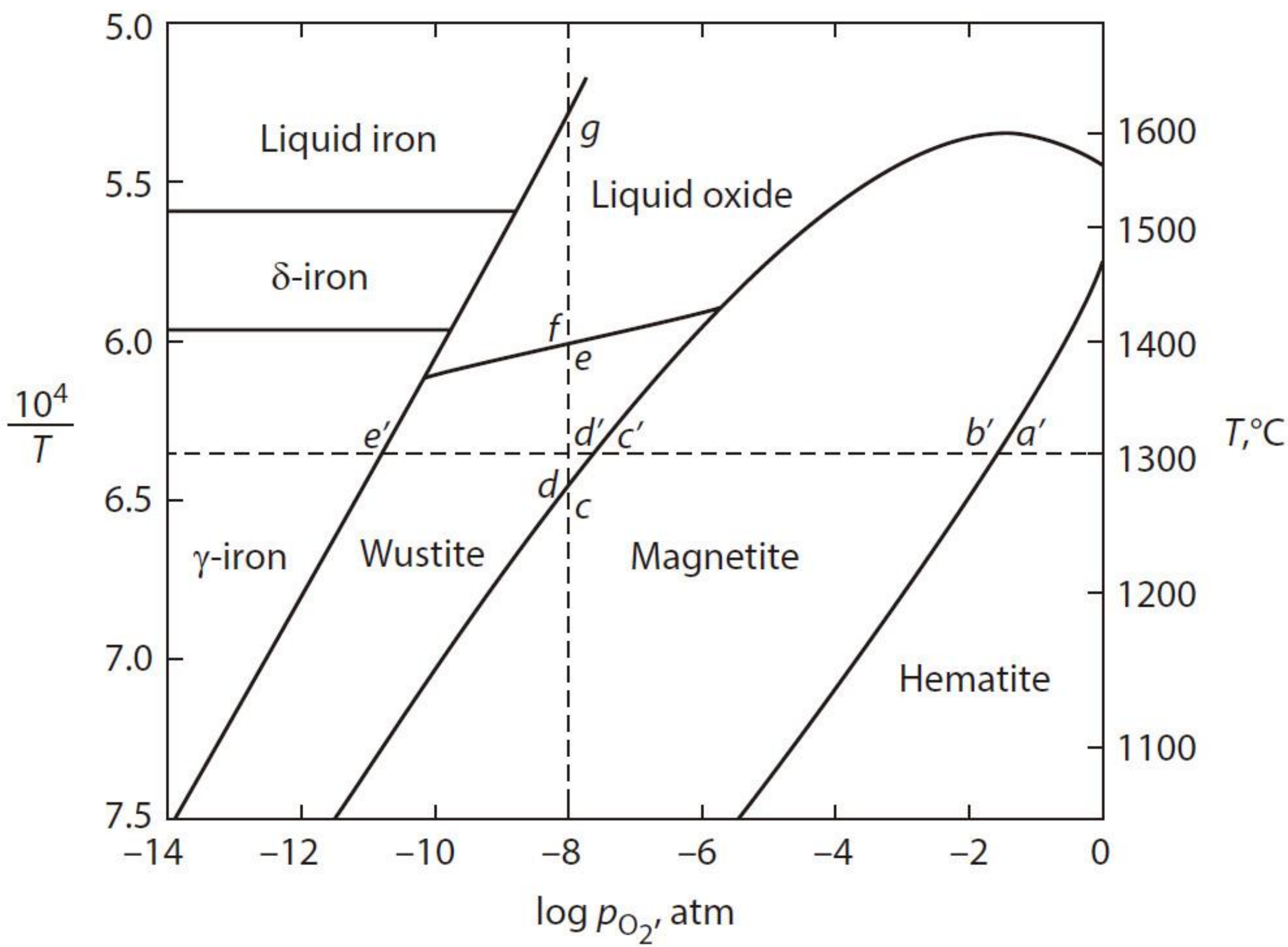

Figure 13.27 Phase stability in the system Fe–Fe_2O_3 as a function of log p_{O_2} and $1/T$.

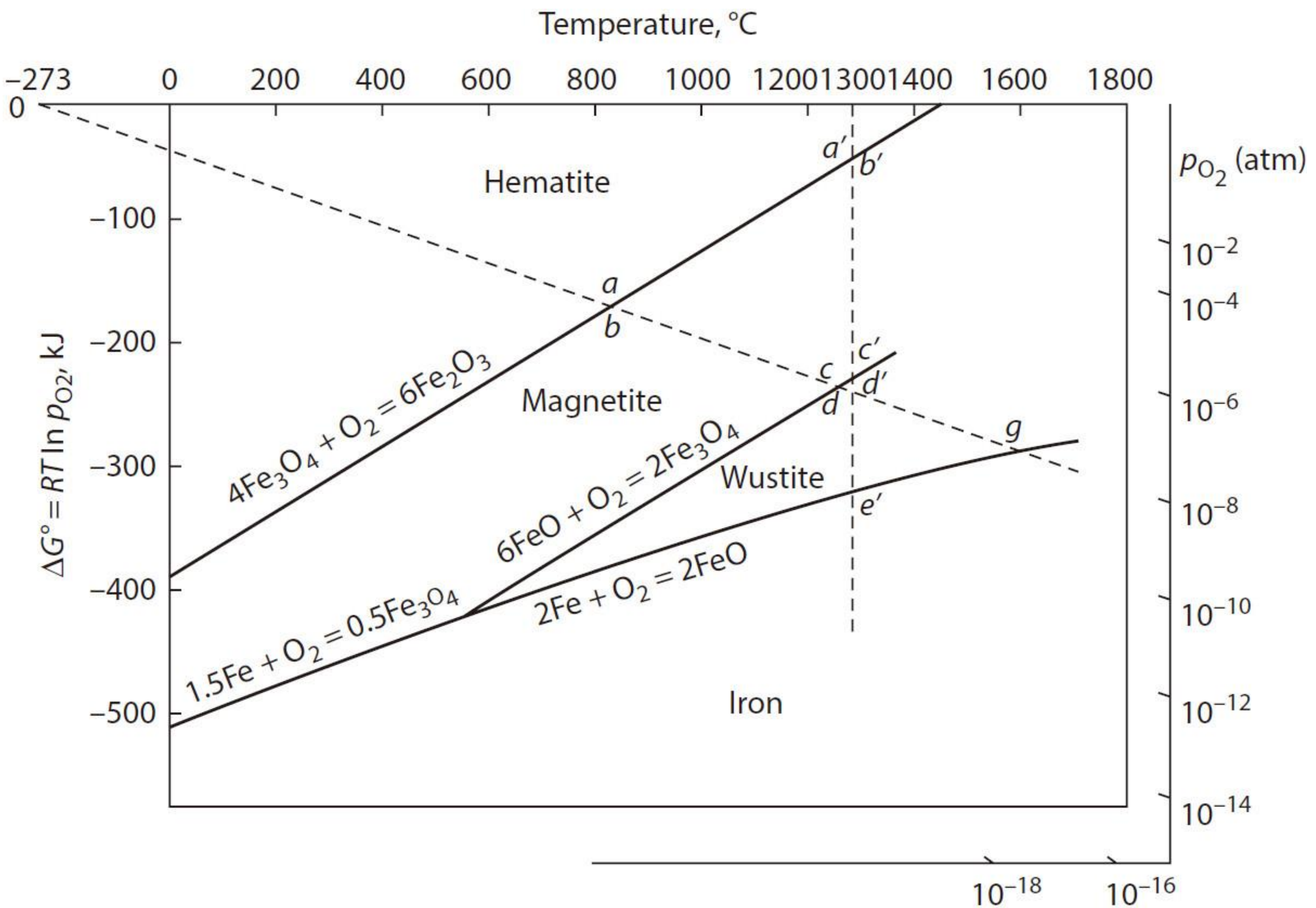

Figure 13.28 Phase stability in the system Fe–Fe_2O_3 as a function of ΔG° and temperature.

13.8 The Formation Of Oxide Phases Of Variable Composition

- ✓ Except for the Fe_3O_4 – Fe_2O_3 line in Figure 13.28, the lines are drawn for oxidation reactions involving the consumption of 1 mole of oxygen—that is, of the type

$$\mathrm{Fe}_x\mathrm{O}_y + \mathrm{O}_2 = \mathrm{Fe}_x\mathrm{O}_{(2+y)}$$

  in which the lower oxide of composition Fe_xO_y is in equilibrium with the higher oxide of composition $Fe_xO_{(2+y)}$.

- ✓ The Fe_3O_4 – Fe_2O_3 line is hypothetical and applies to the stoichiometric compounds (stoichiometric Fe_3O_4 contains Fe at a higher activity than does the composition in equilibrium with hematite).
- ✓ In the Ellingham diagram, lines which radiate from the origin ($\Delta G^\circ = 0$, T = 0 K) are oxygen isobars.
- ✓ The distinct advantage of the Ellingham-type representation of reaction and phase equilibria is its ability to indicate, at a glance, the relative stabilities of a large number of metal–oxygen systems, as was seen in Figure 12.13.

Outlines

13.9 The Solubility Of Gases In Metals

✓ It is invariably found that molecular gases dissolve in metals as atoms. For example, if pure liquid Ag is brought into contact with oxygen gas at a relatively low pressure, the following series of events occur:

1. Molecules of O_2 striking the surface of the liquid Ag become adsorbed on the surface.
2. The adsorbed molecules dissociate to form O atoms adsorbed on the surface.
3. The adsorbed O atoms diffuse from the surface into the bulk melt.

13.9 The Solubility Of Gases In Metals

- ✓ The overall reaction can be written as

$$\frac{1}{2}\mathrm{O}_{2(g)} = [\mathrm{O}]_{\text{in Ag}}$$

and equilibrium is attained when the partial molal Gibbs free energy of oxygen in solution in the liquid Ag, $\bar{G}_{\mathrm{O}}$, is equal to the molar Gibbs free energy of oxygen in the gas phase, $\frac{1}{2} G_{\mathrm{O}_2}$.

- ✓ The standard state for a gas dissolved in a metal can be chosen as the 1 wt% standard state, discussed in Section 13.3, or as the 1 atom percent (at%) standard state, which is the point on the Henry' s law line at a mole fraction of solute of 0.01.

- ✓ Thus, the standard change in the molar Gibbs free energy which occurs when a gaseous species A_2, at 1 atm pressure and the temperature *T*, is dissolved in a metal at the concentration X_A = 0.01 (1 at%), at the temperature *T*, according to

$$\frac{1}{2}\mathrm{A}_{2(g,P=1\ \mathrm{atm})}=[\mathrm{A}]_{(1\ \mathrm{at\%})} \tag{13.15}$$

is

$$\Delta G^{\circ}_{1\ \mathrm{at\%}}=-RT\ln K_{1\ \mathrm{at\%}}=-RT\ln\frac{[h_{\mathrm{A}}]_{1\ \mathrm{at\%}}}{p_{\mathrm{A}_2}^{1/2}} \tag{13.16}$$

in which $[h_A]_{(1\ at\%)}$ is the activity of A in solution in the metal relative to the 1 at% standard state. If the solute obeys Henry’ s law,

$$K_{1\ \mathrm{at\%}}=\frac{[\mathrm{at\%\ A}]}{p_{\mathrm{A}_2}^{1/2}} \tag{13.17}$$

13.9 The Solubility Of Gases In Metals

- ✓ If, however, the 1 wt% standard state is chosen for the solute, then for

$$\frac{1}{2}\mathrm{A}_{2(g,P=1\ \mathrm{atm})} = [\mathrm{A}]_{(1\ \mathrm{wt\%})} \tag{13.18}$$

$$\Delta G^{\circ}_{1\ \mathrm{wt\%}} = -RT\ln K_{1\ \mathrm{wt\%}} = -RT\ln\frac{[h_{\mathrm{A}}]_{1\ \mathrm{wt\%}}}{p_{\mathrm{A}_2}^{1/2}} \tag{13.19}$$

in which $[h_A]_{(1\ \mathrm{wt\%})}$ is the activity of A in solution relative to the 1 wt% standard state.

- ✓ If Henry' s law is obeyed,

$$K_{1\ \mathrm{wt\%}} = \frac{[\mathrm{wt\%\ A}]}{p_{\mathrm{A}_2}^{1/2}} \tag{13.20}$$

13.9 The Solubility Of Gases In Metals

- ✓ For the solution of oxygen in liquid Ag in the range 1213–1573 K,

$$\Delta G^{\circ}_{1\ \text{at\%}} = -14{,}310 + 5.44T \text{ J} \qquad \text{(i)}$$

- ✓ A concentration of 1 wt% O in Ag corresponds to a mole fraction of

$$\frac{\frac{1}{16}}{\frac{1}{16}+\frac{99}{107.9}} = 0.0638$$

and thus, the change in the molar Gibbs free energy for

$$[\mathrm{O}]_{(\text{in Ag. 1 at\%})} = [\mathrm{O}]_{(\text{in Ag. 1 wt\%})}$$

is

$$\Delta G^{\circ}_{(\text{ii})} = -8.3144T \ln\frac{0.0638}{0.01} = -15.40T \text{ J} \qquad \text{(ii)}$$

13.9 The Solubility Of Gases In Metals

- The sum of $\Delta G^\circ_{1\text{ at\%}}$ and $\Delta G^\circ_{(\text{ii})}$ gives

$$\Delta G^\circ_{1\text{ wt\%}} = -14,310 - 9.96T \text{ J} \qquad \text{(iii)}$$

- For the solution of oxygen in solid Ag in the range 573–1173 K,

$$\Delta G^\circ_{1\text{ at\%}} = 49,620 - 15.77T \text{ J} \qquad \text{(iv)}$$

- Oxygen obeys Henry' s law in liquid Ag, and thus, from Equations (i), 13.16, and 13.17, the solubility of oxygen in liquid silver is

$$\text{at\% O} = p_{O_2}^{1/2} \exp\left(\frac{1721}{T} - 0.654\right) \qquad \text{(v)}$$

and, from Equations (iv), 13.16, and 13.17, the solubility of oxygen in solid Ag is

$$\text{at\% O} = p_{O_2}^{1/2} \exp\left(-\frac{5967}{T} + 1.90\right) \qquad \text{(vi)}$$

13.9 The Solubility Of Gases In Metals

- The phase equilibria in the system Ag–O at an oxygen pressure of 1 atm are shown in Figure 13.29.
- Equation (i) gives $\Delta H^{\circ}_{1\ \text{at\%}}$ for the change of state

$$\frac{1}{2}\text{O}_{2(g,P=1\ \text{atm})} = [\text{O}]_{(1\ \text{at\% in liquid Ag})}$$

as –14,310 J.

- Thus, since the enthalpy change is negative, decreasing the temperature causes the equilibrium to shift to the right, with the consequence that, at constant oxygen pressure, the solubility of oxygen in liquid Ag increases with decreasing temperature.
- From Equation (v), the maximum solubility of O in liquid Ag in Figure 13.29 is 2.14 at% at 940 ℃ (the state *b*).

13.9 The Solubility Of Gases In Metals

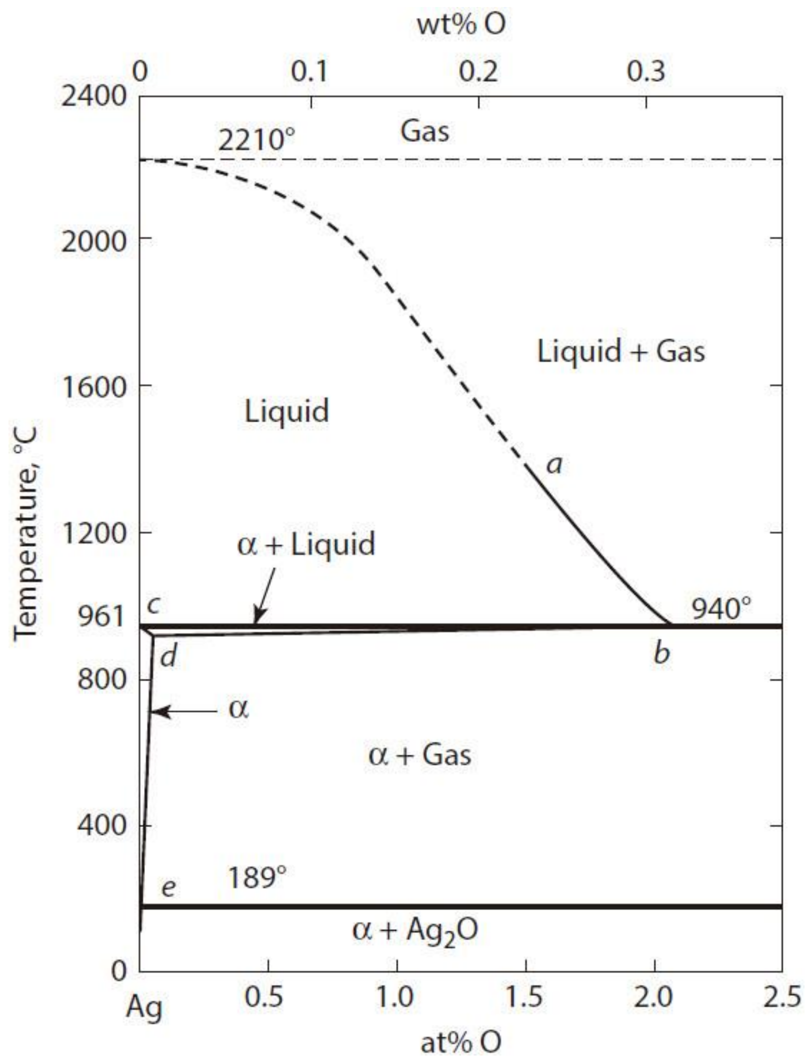

Figure 13.29 Phase equilibria in the system Ag–O at an oxygen pressure of 1 atm.

13.9 The Solubility Of Gases In Metals

- Equation (vi) gives the solubility of O in solid Ag at 940 ℃ as 0.049 at% (the state *d*), and thus, the transformation

$$\mathrm{Ag}_{(l,2.14\ \mathrm{at\%\ O})} \rightarrow \mathrm{Ag}_{(s,0.049\ \mathrm{at\%\ O})} + \mathrm{O}_{2(g,P=1\ \mathrm{atm})}$$

  occurs at 940 ℃.

- The evolution of oxygen during the freezing of oxygen-containing liquid Ag causes the phenomenon of *spitting*, during which droplets of liquid silver are ejected from the freezing mass of liquid.

- Equation (iv) gives $\Delta H^{\circ}_{1\ \mathrm{at\%}}$ for the change of state

$$\frac{1}{2}\mathrm{O}_{2(g,P=1\ \mathrm{atm})} = \mathrm{O}_{1\ \mathrm{at\%\ in\ solid\ Ag}}$$

  as 49,620 J, which being positive, requires that the solubility of O in solid Ag decreases with decreasing temperature.

13.9 The Solubility Of Gases In Metals

- ✓ From Equation (vi), the solubility of O in solid Ag at an oxygen pressure of 1 atm decreases from 0.049 at% at 940 ℃ to 1.7×10^{-5} at% at 189 ℃ (the point *e* in Figure 13.29). ΔG° for the reaction

$$2\mathrm{Ag}_{(s)}+\frac{1}{2}\mathrm{O}_{2(g)}=\mathrm{Ag}_2\mathrm{O}_{(s)}$$

is written as

$$\Delta G^\circ=-30{,}540+66.11T\ \mathrm{J}$$

- ✓ The temperature at which $p_{\mathrm{O}_{2,\mathrm{eq}}}=1\ \mathrm{atm}$ is thus 30,540/66.11 = 462 K (189 ℃), and the three-phase invariant equilibrium is shown at 189 ℃ in Figure 13.29.

- ✓ The solubility of oxygen in Ag is shown as a function of temperature and oxygen pressure in Figure 13.30.

13.9 The Solubility Of Gases In Metals

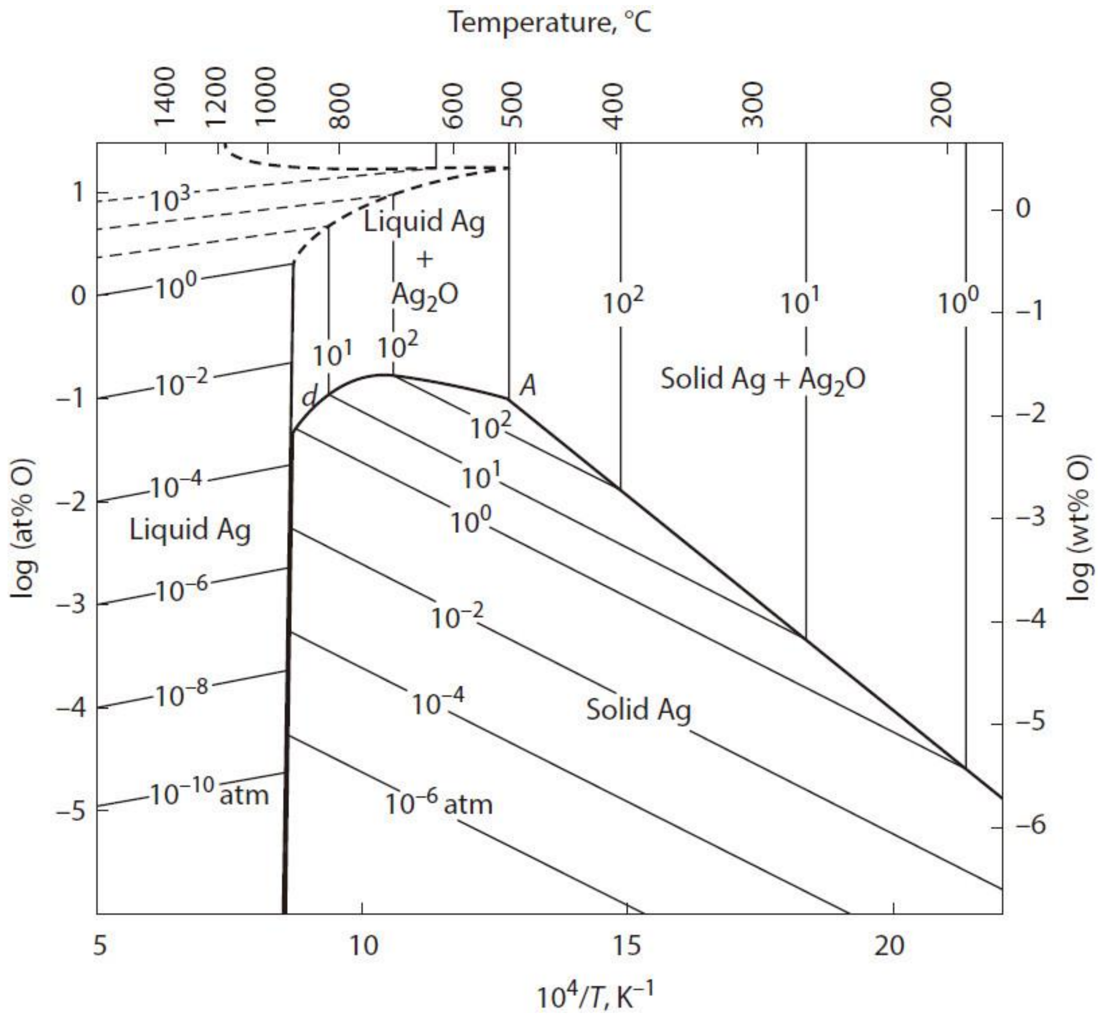

Figure 13.30 The solubility of oxygen in silver as a function of oxygen pressure and temperature.

13.9 The Solubility Of Gases In Metals

- ✓ On a plot of log (at% O) versus inverse temperature, the oxygen isobars in a single-phase field are parallel lines, the slopes of which are determined by the molar enthalpies of solution of oxygen.
- ✓ From Equation (v), the slope of the lines in the phase field of liquid silver is 1721, and from Equation (vi), the slope in the phase field of solid silver is –5967.
- ✓ The points *b* and *d* on the 1 atm isobar at 940 ℃ correspond with the points *b* and *d* in Figure 13.29.
- ✓ Increasing the oxygen pressure decreases the temperature at which the equilibrium involving liquid Ag, solid Ag, and O_2 gas occurs and increases the temperature at which the equilibrium involving solid Ag, Ag_2O, and O_2 gas occurs.

13.9 The Solubility Of Gases In Metals

- ✓ The two temperatures coincide at 508 ℃ when the oxygen pressure is 414 atm (the state *A* in Figure 13.30). This four-phase equilibrium in a binary system has zero degrees of freedom.
- ✓ Recasting Equation 13.17 as

$$\left[\text{at\% A}\right] = k(T)\, p_{\mathrm{A}_2}^{1/2} \tag{13.21}$$

gives an equation known as *Sieverts' law* (Adolf Sieverts, 1874–1947) and the temperature-dependent constant in Equation 13.21, $k(T)$, which is known as *Sieverts' constant*, is evaluated as the concentration of A in the metal equilibrated, at the temperature T, with gaseous A_2 at 1 atm pressure.

- ✓ Sieverts measured the solubility of oxygen in liquid Ag in 1907.

Outlines

13.10 Solutions Containing Several Dilute Solutes

- ✓ The behavior of a dilute solute in a binary solution is determined by the nature and magnitude of the interactions between the solute and solvent atoms.
- ✓ However, when a second dilute solute is added, three types of interaction occur—namely, solvent–solute I, solvent–solute II, and solute I–solute II—and the thermodynamic behavior of the system is determined by the relative magnitudes of the three types of interaction.
- ✓ Consider the exposure of liquid iron to a gaseous mixture of hydrogen and oxygen.
- ✓ The equilibrium

$$\mathrm{H}_{2(g)} + \frac{1}{2}\mathrm{O}_{2(g)} = \mathrm{H_2O}_{(g)} \qquad \text{(i)}$$

is established in the gas phase, and thus,

$$p_{\mathrm{H}_2} p_{\mathrm{O}_2}^{1/2} = \frac{p_{\mathrm{H_2O}}}{K_{(\mathrm{i})}}$$

13.10 Solutions Containing Several Dilute Solutes

- ✓ Since both hydrogen and oxygen have some limited solubility in liquid iron, both gases will dissolve atomically until their respective activities in the iron, with respect to the 1 atm pressure standard state, equal the respective partial pressures in the gas phase.
- ✓ Alternatively, with respect to the 1 wt% in Fe standard state,

$$\frac{1}{2}O_{2(g)} = [O]_{(1\text{ wt\% in Fe})} \qquad \text{(ii)}$$

and

$$\frac{1}{2}H_{2(g)} = [H]_{(1\text{ wt\% in Fe})} \qquad \text{(iii)}$$

for which

$$h_{O(1\text{ wt\%})} = K_{(ii)}\,p_{O_2}^{1/2}$$

and

$$h_{H(1\text{ wt\%})} = K_{(iii)}\,p_{H_2}^{1/2}$$

13.10 Solutions Containing Several Dilute Solutes

- ✓ Equilibrium in the metal phase is given as

$$h_{\mathrm{H(1\ wt\%)}}^{2} h_{\mathrm{O(1\ wt\%)}} = \frac{K_{\mathrm{(iii)}}^{2} K_{\mathrm{(ii)}}}{K_{\mathrm{(i)}}} p_{\mathrm{H_2O}}$$

or

$$f_{\mathrm{H(1\ wt\%)}}^{2} f_{\mathrm{O(1\ wt\%)}} [\mathrm{wt\%\ H}]^{2} [\mathrm{wt\%\ O}] = \frac{K_{\mathrm{(iii)}}^{2} K_{\mathrm{(ii)}}}{K_{\mathrm{(i)}}} p_{\mathrm{H_2O}}$$

- ✓ The solubilities of H and O (expressed as weight percentages) are thus determined by the values of the activity coefficients ($f_{\mathrm{i(1\ wt\%)}}$) of H and O, and the questions to be answered are
  1. How is the activity coefficient of O in Fe influenced by the presence of H?
  2. How is the activity coefficient of H in Fe influenced by the presence of O?

13.10 Solutions Containing Several Dilute Solutes

- ✓ This problem is dealt with by the introduction of interaction coefficients and interaction parameters.
- ✓ In the binary A–B, the activity of B in dilute solution with respect to the Henrian standard state is given by

$$h_B = f_B^B X_B$$

- ✓ If, holding the concentration of B constant, the addition of a small amount of C changes the value of the activity coefficient of B to f_B, then the difference between f_B and f_B^B is quantified by the expression

$$f_B = f_B^B f_B^C \tag{13.22}$$

where f_B^C is called the *interaction coefficient* of C on B and is a measure of the effect, on the behavior of B, of the presence of a specific concentration of C, at the same concentration of B.

13.10 Solutions Containing Several Dilute Solutes

- ✓ Similarly, if a small amount of D is added to the A–B solution, as a result of which the value of the activity coefficient of B changes from f_B^B to f_B, then

$$f_B = f_B^B f_B^D$$

- ✓ Now consider the system A–B–C–D.
- ✓ Mathematical analysis of such a system is possible only if f_B^D is independent of the concentration of C and if f_B^C is independent of the concentration of D.
- ✓ Consider that the interaction coefficient of the solute *i* on the solute *j* is independent of the other solutes present, in which case the interaction coefficients may be combined by means of a Taylor' s expansion of $\ln f_i$ as a function of the concentrations of the solutes.

13.10 Solutions Containing Several Dilute Solutes

- ✓ For example, for the binary system A–B in which A is the solvent,

$$\ln f_{\mathrm{B}} = \text{some function of the mole fraction of B}$$

$$= \ln f_{\mathrm{B}}^{\circ} + \left(\frac{\partial \ln f_{\mathrm{B}}}{\partial X_{\mathrm{B}}}\right) X_{\mathrm{B}} + \frac{1}{2}\left(\frac{\partial^2 \ln f_{\mathrm{B}}}{\partial X_{\mathrm{B}}^2}\right) X_{\mathrm{B}}^2 + \cdots$$

- ✓ In this expression, the partial derivatives are the limiting values reached as $X_{\mathrm{B}} \to 0$.
- ✓ For the multicomponent system A–B–C–D,

$$\ln f_{\mathrm{B}} = \text{some function of the fractions of B, C and D}$$

$$= \ln f_{\mathrm{B}}^{\circ} + \left(\frac{\partial \ln f_{\mathrm{B}}}{\partial X_{\mathrm{B}}} X_{\mathrm{B}} + \frac{\partial \ln f_{\mathrm{B}}}{\partial X_{\mathrm{C}}} X_{\mathrm{C}} + \frac{\partial \ln f_{\mathrm{B}}}{\partial X_{\mathrm{D}}} X_{\mathrm{D}}\right)$$

$$+ \left(\frac{1}{2}\frac{\partial^2 \ln f_{\mathrm{B}}}{\partial X_{\mathrm{B}}^2} X_{\mathrm{B}}^2 + \frac{\partial^2 \ln f_{\mathrm{B}}}{\partial X_{\mathrm{B}} \partial X_{\mathrm{C}}} X_{\mathrm{B}} X_{\mathrm{C}} + \cdots\right)$$

in which, again, the partial derivatives are the limiting values as the mole fractions of the solutes approach zero.

13.10 Solutions Containing Several Dilute Solutes

- At very low concentration, the terms containing the products of mole fractions are small enough to be ignored, and also, choice of the Henrian standard state makes $f_B^\circ = 1$.
- Thus,

$$\ln f_B = \frac{\partial \ln f_B}{\partial X_B} X_B + \frac{\partial \ln f_B}{\partial X_C} X_C + \frac{\partial \ln f_B}{\partial X_D} X_D \qquad (13.23)$$

$$= \varepsilon_B^B X_B + \varepsilon_B^C X_C + \varepsilon_B^D X_D$$

where

$$\varepsilon_j^i = \left.\frac{\partial \ln f_j}{\partial X_i}\right|_{X_i \to 0}$$ 这个公式和课本不同

is called the *interaction parameter* of *i* on *j* and is obtained as the limiting slope of a plot of lnf_j against X_i at constant X_j.

13.10 Solutions Containing Several Dilute Solutes

✓ ε_j^i and ε_j^j are related as follows. For the general system

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$$\frac{\partial^2 G}{\partial n_i \partial n_j} = \frac{\partial \bar{G}_i}{\partial n_j} = \frac{\partial \bar{G}_j}{\partial n_i}$$

and since $\partial \bar{G}_i = RT\partial \ln a_i = RT\partial \ln f_i$, then

$$\frac{\partial \ln f_j}{\partial n_i} = \frac{\partial \ln f_i}{\partial n_j}$$

and thus,

$$\varepsilon_i^j = \varepsilon_j^i \qquad (13.24)$$

13.10 Solutions Containing Several Dilute Solutes

- It is often more convenient to consider the concentrations of the solutes in terms of weight percentages and to use logarithms to the base 10, in which case Equation 13.23 becomes

$$\log f_{\mathrm{B}} = \frac{\partial \log f_{\mathrm{B}}}{\partial \text{ wt\% B}} \text{wt\% B} + \frac{\partial \log f_{\mathrm{B}}}{\partial \text{ wt\% C}} \text{wt\% C} + \frac{\partial \log f_{\mathrm{B}}}{\partial \text{ wt\% D}} \text{wt\% D} \quad (13.25)$$

$$\log f_{\mathrm{B}} = e_{\mathrm{B}}^{\mathrm{B}} \text{wt\% B} + e_{\mathrm{B}}^{\mathrm{C}} \text{wt\% C} + e_{\mathrm{B}}^{\mathrm{D}} \text{wt\% D}$$

- Multiplying Equation 13.25 by 2.303 and comparing, term by term, with Equation 13.23 gives

$$\varepsilon_{\mathrm{B}}^{i} X_i = 2.303\, e_{\mathrm{B}}^{i} \text{ wt\% } i$$

and since, at small concentrations of B and *i*,

$$X_i \approx \frac{\text{wt\% } i \cdot \mathrm{MW_A}}{100 \mathrm{MW}_i}$$

13.10 Solutions Containing Several Dilute Solutes

then

$$e_{\mathrm{B}}^{i} = \frac{1}{230.3}\frac{\mathrm{MW_A}}{\mathrm{MW}_i}\varepsilon_{\mathrm{B}}^{i}$$

and

$$e_{\mathrm{B}}^{i} = \frac{\mathrm{MW_B}}{\mathrm{MW}_i}e_{j}^{\mathrm{B}}$$

✓ Pehlke and Elliott* have determined that nitrogen, dissolved in liquid iron at 1600 ℃, obeys Sieverts' law according to

$$[\mathrm{wt\%\ N}] = kp_{\mathrm{N_2}}^{1/2} \tag{iv}$$

where k = 0.045 at 1873 K, and they have measured the effects of the presence of a second dilute solute on the thermodynamics of nitrogen dissolved in liquid iron.

13.10 Solutions Containing Several Dilute Solutes

- ✓ These systems are particularly amenable to experimental study because of the ease with which the activity of nitrogen can be controlled by the gas phase.
- ✓ The interaction parameters, e_{N}^{i}, are determined by maintaining the nitrogen in the melt at constant activity and measuring the variation in the solubility of nitrogen with concentration of the second solute.
- ✓ Since nitrogen in liquid iron obeys Henry' s (Sieverts') law, $f_{\mathrm{N}}^{\mathrm{N}} = 1$, and thus, $e_{\mathrm{N}}^{\mathrm{N}}$ (and $\varepsilon_{\mathrm{N}}^{\mathrm{N}}$) are zero. Consequently, the first terms in Equations 13.23 and 13.25 are zero.
- ✓ If the addition of a second solute X to the Fe–N binary (equilibrated with a fixed p_{N_2}) causes a change in the dissolved nitrogen content from $[\text{wt\% N}]_{\text{Fe–N}}$ to $[\text{wt\% N}]_{\text{Fe–N–X}}$.

13.10 Solutions Containing Several Dilute Solutes

- Then, from Equation (iv),

$$k = \frac{[\text{wt\% N}]_{\text{in Fe-N}}}{p_{\mathrm{N_2}}^{1/2}} = \frac{f_{\mathrm{N}}^{\mathrm{X}}[\text{wt\% N}]_{\text{in Fe-N-X}}}{p_{\mathrm{N_2}}^{1/2}}$$

and thus, $f_{\mathrm{N}}^{\mathrm{X}}$ is obtained experimentally as

$$f_{\mathrm{N}}^{\mathrm{X}} = \left(\frac{[\text{wt\% N}]_{\text{in Fe-N}}}{[\text{wt\% N}]_{\text{in Fe-N-X}}}\right)_{T,\,p_{\mathrm{N_2}}}$$

- The variation of [wt% N] with [wt% X] is shown for several second solutes in Figure 13.31, and the corresponding variation of $\log f_{\mathrm{N}}^{\mathrm{X}}$ with [wt% X] is shown in Figure 13.32. The values of $e_{\mathrm{N}}^{\mathrm{X}}$ are obtained as the slopes of the linear portions of the lines in Figure 13.32.

13.10 Solutions Containing Several Dilute Solutes

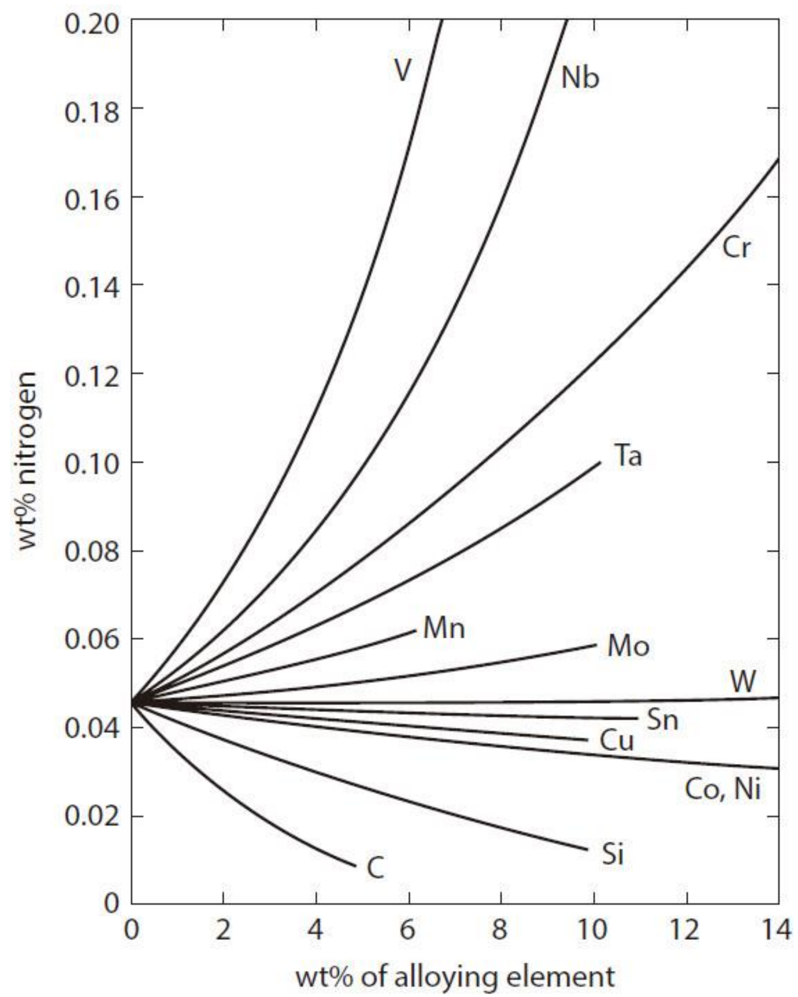

Figure 13.31 The effect of alloying elements on the solubility of nitrogen at 1 atm pressure in liquid binary iron alloys at 1600°C.

13.10 Solutions Containing Several Dilute Solutes

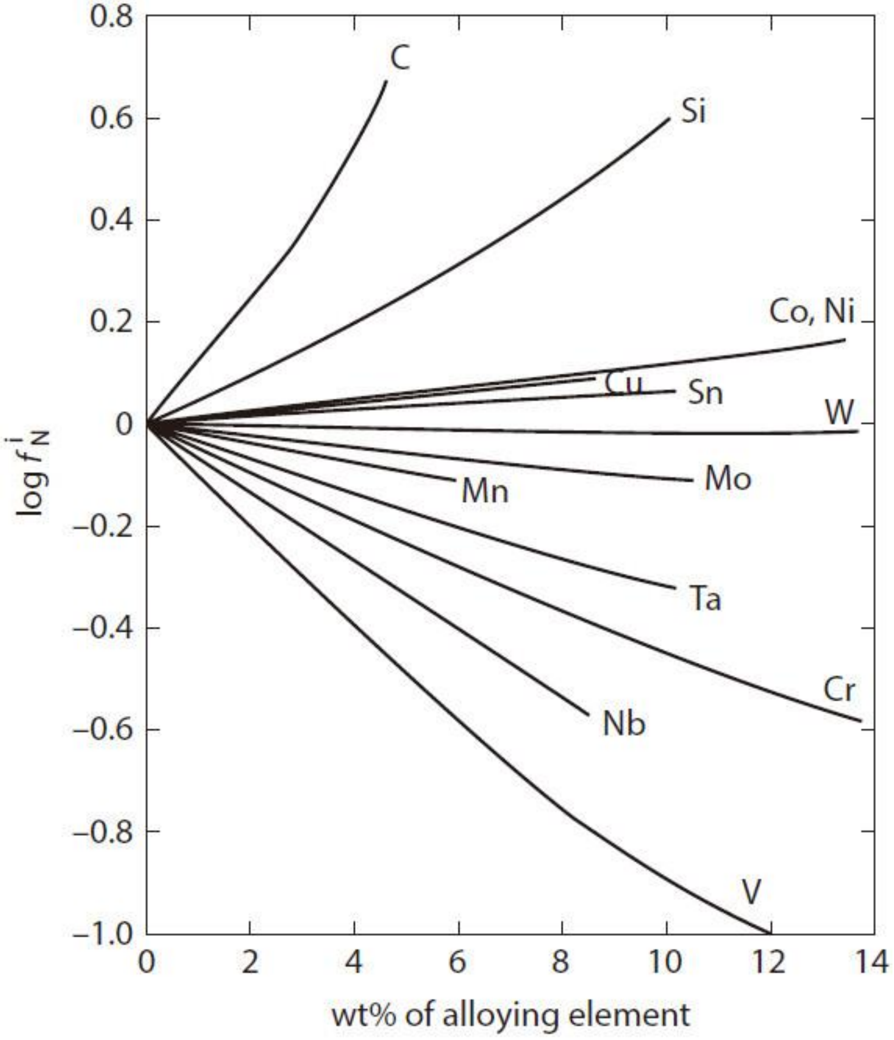

Figure 13.32 The activity coefficients of nitrogen in binary iron alloys at 1600°C.

13.10 Solutions Containing Several Dilute Solutes

- ✓ Thus, in a multicomponent liquid iron alloy containing several solutes including nitrogen, if the effect of any one solute on f_N is independent of the presence of any other solute, then the total effect of the solutes on f_N is the sum of their individual effects, and if $\log f_N^X$ is a linear function of [wt% X], then f_N is given by Equation 13.25.
- ✓ If, however, the concentrations of X are higher than the limits of linear variation of $\log f_N^X$ with [wt% X], then a graphical solution is required.
- ✓ In these cases, the value of $\log f_N^X$ for each value of [wt% X] is read from the graph (Figure 13.32), and $\log f_N$ is obtained as

$$\log f_N = \sum_X \log f_N^X$$

or

$$f_N = \prod_X f_N^X$$

13.10 Solutions Containing Several Dilute Solutes

- ✓ Figure 13.32 indicates that, as a general rule, e_N^X is a negative quantity when X forms a nitride which is more stable than iron nitride, and that the order of increasing magnitude of $\left|e_N^X\right|$ follows the order of increasing magnitude of the Gibbs free energy of formation of the nitride of X.

- ✓ Similarly, e_N^X is a positive quantity when X has a greater affinity for iron than either X has for N or iron has for nitrogen.

- ✓ The values of the interaction coefficients for several elements in dilute solution in iron at 1600 ℃ are listed in Table 13.1 .

13.10 Solutions Containing Several Dilute Solutes

Table 13.1 Interaction Coefficients for Dilute Solutions of Elements Dissolved in Liquid Iron at 1600°C

Element(*i*)	Element (*j*)											
	Al	**C**	**Co**	**Cr**	**H**	**Mn**	**N**	**Ni**	**O**	**P**	**S**	**Si**
Al	4.8	11	—	—	(34)	—	(0.5)	—	−160	—	4.9	6
C	(4.8)	22	1.2	−2.4	(72)	—	(11.1)	1.2	(−9.7)	—	9	10
Co	—	(6)	—	—	(11)	—	(4.7)	—	(2.6)	—	—	—
Cr	—	(−10)	—	—	(−11)	—	(−16.6)	—	(−13)	—	(−3.55)	—
H	1.3	6.0	0.18	−0.22	0	−0.14	—	0	—	1.1	0.8	2.7
Mn	—	—	—	—	(−7.7)	—	(−7.8)	—	(0)	—	(−4.3)	(0)
N	0.3	13	1.1	−4.5	—	−2	0	1	5.0	5.1	1.3	4.7
Ni	—	(5.9)	—	—	(0)	—	(4.2)	0	(2.1)	—	(0)	(1.0)
O	−94	−13	0.7	−4.1	—	0	(5.7)	0.6	−20	7.0	−9.1	−14
P	—	—	—	—	(34)	—	(11.3)	—	(13.5)	—	(4.3)	(9.5)
S	5.8	(24)	—	−2.2	(26)	−2.5	(3.0)	0	(−18)	4.5	−2.8	6.6
Si	(6.3)	24	—	—	(76)	0	(9.3)	0.5	(−25)	8.6	(5.7)	32

Some interaction coefficients $e_i^j \times 10^2$ for dilute solutions of elements dissolved in liquid iron at 1600°C. Values in parentheses are calculated from $e_i^j = (MW_i / MW_j) e_j^i$. (From J. F. Elliott, M. Gleiser, and V. Ramakrishna, *Thermochemistry for Steelmaking*, vol. 2, Addison-Wesley, Reading, MA, 1963.)

13.10 Solutions Containing Several Dilute Solutes

Example 1

In view of the introduction of interaction parameters, the example of the Si-O equilibrium in liquid Fe, discussed in Section 13.3, can now be reexamined. In this example, it was determined that, for

$$\mathrm{Si}_{(1\text{ wt\% in Fe})} + \mathrm{O}_{2(g)} = \mathrm{SiO}_{2(s)}$$

$$\Delta G^\circ = -833,400 + 229.5T \text{ J}$$

For

$$\frac{1}{2}\mathrm{O}_{2(g)} = [\mathrm{O}]_{(1\text{ wt\% in Fe})}$$

$$\Delta G^\circ = -111,300 - 6.41T \text{ J} \qquad \text{(i)}$$

13.10 Solutions Containing Several Dilute Solutes

and thus, for

$$\mathrm{Si}_{(1\ \mathrm{wt\%\ in\ Fe})} + 2\mathrm{O}_{(1\ \mathrm{wt\%\ in\ Fe})} = \mathrm{SiO}_{2(s)}$$

$$\Delta G^{\circ} = -610{,}800 + 242.32T\ \mathrm{J} \qquad \text{(ii)}$$

From Equation (i) at 1600 ℃,

$$\frac{h_{\mathrm{O}(1\ \mathrm{wt\%})}}{p_{\mathrm{O}_2}^{1/2}} = 2.746 \times 10^3 \qquad \text{(iii)}$$

and, from Equation (ii) at 1600 ℃,

$$\frac{a_{\mathrm{SiO}_2}}{h_{\mathrm{Si}(1\ \mathrm{wt\%})} h_{\mathrm{O}(1\ \mathrm{wt\%})}^2} = 2.380 \times 10^4 \qquad \text{(iv)}$$

13.10 Solutions Containing Several Dilute Solutes

Thus, with $p_{O_2} = 5.57 \times 10^{-12}$ atm and $a_{SiO_2} = 1$, Equation (iii) gives

$$h_{O(1\ \mathrm{wt\%})} = 6.48 \times 10^{3} \qquad \text{(v)}$$

and Equation (iv) gives

$$h^2_{O(1\ \mathrm{wt\%})} h_{Si(1\ \mathrm{wt\%})} = 4.0 \times 10^{-5} \qquad \text{(vi)}$$

Division of Equation (vi) by $h^2_{O(1\ \mathrm{wt\%})}$ from Equation (v) gives

$$h_{Si(1\ \mathrm{wt\%})} = 1 \qquad \text{(vii)}$$

In the previous treatment the assumption that Si obeys Henry' s law leads to the conclusion that

$$h_{Si(1\ \mathrm{wt\%})} = \mathrm{wt\%\ Si} = 1$$

13.10 Solutions Containing Several Dilute Solutes

At 1600 ℃, from Table 13.1,

$$e_{\mathrm{O}}^{\mathrm{Si}} = -0.14 \qquad e_{\mathrm{O}}^{\mathrm{O}} = -0.2$$

$$e_{\mathrm{Si}}^{\mathrm{O}} = -0.25 \qquad e_{\mathrm{Si}}^{\mathrm{Si}} = -0.32$$

Thus, from Equation (v),

$$\log f_{\mathrm{O}} + \log[\mathrm{wt\%\ O}] = \log\left(6.48\times10^{-3}\right)$$

or

$$-0.2\times[\mathrm{wt\%\ O}] - 0.14\times[\mathrm{wt\%\ Si}] + \log[\mathrm{wt\%\ O}] = -2.188 \qquad \text{(viii)}$$

and from Equation (vii),

$$\log f_{\mathrm{Si}} + \log[\mathrm{wt\%\ Si}] = \log(1)$$

or

$$0.32[\mathrm{wt\%\ Si}] - 0.25[\mathrm{wt\%\ O}] + \log[\mathrm{wt\%\ Si}] = 0 \qquad \text{(ix)}$$

13.10 Solutions Containing Several Dilute Solutes

Computer solution of Equations (viii) and (ix) gives

$$[\text{wt\% Si}] = 0.631 \text{ and } [\text{wt\% O}] = 0.00798$$

In the example in Section 13.3, in which the effect of dissolved oxygen was ignored and it was assumed that f_{Si} = 1, the equilibrium weight percentage of Si in iron when $a_{SiO_2} = 1$and $p_{O_2} = 5.57 \times 10^{-12}$atm was 1.0.

It is of interest to determine which of the two initial assumptions, that (1) $e_{Si}^{Si} = 0$ and (2) $e_{Si}^{O} = e_{O}^{Si} = 0$, contributes more to the error in the initial calculation.

Use $e_{Si}^{Si} = 0.32$ and assume that e_{Si}^{O} and e_{O}^{Si} are zero.

13.10 Solutions Containing Several Dilute Solutes

From Equation (ix):

$$0.32\left[\text{wt\% Si}\right]+\log\left[\text{wt\% Si}\right]=0$$

which gives [wt% Si] = 0.629, and from Equation (viii),

$$-0.20\times\left[\text{wt\% O}\right]-0.24\times 0.629+\log\left[\text{wt\% O}\right]=-2.188$$

which gives [wt% O] = 0.00797.

The error introduced by ignoring the interaction between Si and O in solution in Fe is thus seen to be negligible in comparison with that introduced by assuming that Si obeys Henry’ s law over some initial range of composition.

13.10 Solutions Containing Several Dilute Solutes

Example 2 **(self study)**

Calculate the equilibrium oxygen content of an Fe–C–O alloy which, at 1600 ℃, contains 1 wt% C and is under a pressure of 1 atm of CO.

$$\text{For}\quad \mathrm{C}_{(gr)} + \frac{1}{2}\mathrm{O}_{2(g)} = \mathrm{CO}_{(g)},\ \Delta G^{\circ} = -111,700 - 87.65T\ \mathrm{J}$$

$$\text{For}\quad \mathrm{C}_{(gr)} = \mathrm{C}_{(1\ \mathrm{wt\%\ in\ Fe})},\quad \Delta G^{\circ} = 22,600 - 42.26T\ \mathrm{J}$$

$$\text{For}\quad \frac{1}{2}\mathrm{O}_{2(g)} = \mathrm{O}_{(1\ \mathrm{wt\%\ in\ Fe})},\quad \Delta G^{\circ} = -111,300 - 6.41T\ \mathrm{J}$$

Thus, for

$$\mathrm{C}_{(1\ \mathrm{wt\%})} + \mathrm{O}_{(1\ \mathrm{wt\%})} = \mathrm{CO}_{(g)},\ \Delta G^{\circ} = -23,000 - 38.98T\ \mathrm{J}$$

Therefore,

$$\Delta G^{\circ}_{1873\ \mathrm{K}} = -96,010\ \mathrm{J}$$

and

$$\frac{p_{\mathrm{CO}}}{h_{\mathrm{C}} h_{\mathrm{O}}} = 476$$

13.10 Solutions Containing Several Dilute Solutes

Thus,

$$h_{\mathrm{C}} h_{\mathrm{O}} = f_{\mathrm{C}}[\mathrm{wt\%\ C}]\, f_{\mathrm{O}}[\mathrm{wt\%\ O}] = 2.1\times 10^{-3}\, p_{\mathrm{CO}}$$

At 1600 ℃,

$$e_{\mathrm{C}}^{\mathrm{C}} = 0.22 \qquad e_{\mathrm{O}}^{\mathrm{O}} = -0.2$$

$$e_{\mathrm{C}}^{\mathrm{O}} = -0.097 \qquad e_{\mathrm{O}}^{\mathrm{C}} = -0.13$$

Thus, for 1 wt% C and p_{CO} = 1 atm,

$$\log[\mathrm{wt\%\ O}] - 0.297[\mathrm{wt\%\ O}] = -2.768$$

solution of which gives [wt% O] = 0.00171.

If all of the interaction parameters had been ignored, the weight percentage of O would have been calculated as 0.00210.

13.10 Solutions Containing Several Dilute Solutes

Example 3 **(self study)**

The partial pressure of hydrogen in the atmosphere is such that an Fe–C–Ti melt containing 1 wt% C and 3 wt% Ti contains 5 parts per million (by weight) of hydrogen at 1600 ℃. Calculate the vacuum which is required to decrease the hydrogen content of the melt to 1 ppm, given that $e_{\mathrm{H}}^{\mathrm{Ti}} = -0.08$, that $e_{\mathrm{H}}^{\mathrm{C}} = 0.06$, and that hydrogen in pure iron obeys Henry' s law up to a solubility of 0.0027 wt% under a pressure of 1 atm of hydrogen at 1600 ℃.

For the equilibrium between gaseous hydrogen and dissolved H, given as

$$\tfrac{1}{2}\mathrm{H}_{2(g)} = [\mathrm{H}]_{(1\ \mathrm{wt\%\ in\ Fe})}$$

$$K = \frac{f_{\mathrm{H}(1\ \mathrm{wt\%})}[\mathrm{wt\%\ H}]}{p_{\mathrm{H}_2}^{1/2}}$$

13.10 Solutions Containing Several Dilute Solutes

In pure iron, as H obeys Henry' s law, $f_{\mathrm{H(1\ wt\%)}}$ = 1, and thus,

$$K_{1873\ \mathrm{K}} = 0.0027$$

Thus,

$$\log f_{\mathrm{H(1\ wt\%)}} + \log[\mathrm{wt\%\ H}] - \frac{1}{2}\log p_{\mathrm{H_2}} = \log 0.0027$$

But

$$\log f_{\mathrm{H(1\ wt\%)}} = e_{\mathrm{H}}^{\mathrm{H}}[\mathrm{wt\%\ H}] + e_{\mathrm{H}}^{\mathrm{Ti}}[\mathrm{wt\%\ Ti}] + e_{\mathrm{H}}^{\mathrm{C}}[\mathrm{wt\%\ C}]$$

As $f_{\mathrm{H(1\ wt\%)}}$ = 1, $e_{\mathrm{H}}^{\mathrm{H}} = 0$, and hence, at 1600 ℃,

$$e_{\mathrm{H}}^{\mathrm{Ti}}[\mathrm{wt\%\ Ti}] + e_{\mathrm{H}}^{\mathrm{C}}[\mathrm{wt\%\ C}] + \log[\mathrm{wt\%\ H}] - \frac{1}{2}\log p_{\mathrm{H_2}} = \log 0.0027$$

When [wt% H] = 5×10^{-4},

$$\log p_{H_2} = 2\times\left[(-0.08\times3)+(0.06\times1)+\log\left(5\times10^{-4}\right)-\log 0.0027\right] = -1.825$$

which gives $p_{H_2} = 0.015$ atm.

Similarly, when [wt% H] = 1×10^{-4}, $p_{H_2} = 6\times10^{-4}$ atm.

Thus,

$$[\text{wt\% H}] = 5 \text{ ppm} \qquad \text{when } p_{H_2} = 0.015 \text{ atm} \qquad \text{and } P_{(\text{total})} = 1 \text{ atm}$$

and so

$$[\text{wt\% H}] = 1 \text{ ppm} \qquad \text{when } p_{H_2} = 0.0006 \text{ atm}$$

and

$$P_{(\text{total})} = \frac{0.0006}{0.015} = 0.04 \text{ atm}$$

Thus, in order to achieve the desired decrease in the content of dissolved H, the total pressure must be decreased from 1 to 0.04 atm.

Homework

1. An Fe–Mn solid solution containing $X_{Mn} = 0.001$ is in equilibrium with an FeO–MnO solid solution and an oxygen-containing gaseous atmosphere at 1000 K. How many degrees of freedom does the equilibrium have? What is the composition of the equilibrium oxide solution, and what is the partial pressure of oxygen in the gas phase? Assume that both solid solutions are Raoultian in their behavior.

Homework

2. The elements A and B, which are both solid at 1000 ℃, form two stoichiometric compounds A_2B and AB_2, which are also both solid at 1000 ℃. The system A–B does not contain any solid solutions. A has an immeasurably small vapor pressure at 1000 ℃, and, for the change of state $B_{(s)} = B_{(v)}$,

$$\Delta G^\circ = 187{,}200 - 108.80T \text{ J}$$

The vapor pressure exerted by an equilibrated AB_2–A_2B mixture is given by

$$\log p(\text{atm}) = -\frac{11{,}242}{T} + 6.53$$

and the vapor pressure exerted by an equilibrated A–A_2B mixture is given by

$$\log p(\text{atm}) = -\frac{12{,}603}{T} + 6.9$$

From these data, calculate the standard Gibbs free energies of formation of A_2B and AB_2.

Homework

3. For the change of standard state $V_{(s)} = V_{(1\ \mathrm{wt\%\ in\ Fe})}$

$$\Delta G^\circ = -15,480 - 45.61T\ \text{J}$$

Calculate the value of p_{O_2} at 1600 ℃. If a liquid Fe–V solution is equilibrated with pure solid VO and a gas containing $p_{O_2} = 4.72 \times 10^{-10}$ atm, calculate the activity of V in the liquid solution

(a) with respect to solid V as the standard state,

(b) with respect to liquid V as the standard state,

(c) with respect to the Henrian standard state, and

(d) with respect to the 1 wt% in iron standard state.

The End

13.11 Summary

1. Reaction equilibrium in the reaction

$$aA + bB = cC + dD$$

is established when the reaction has proceeded to the extent that

$$a\overline{G}_A + b\overline{G}_B = c\overline{G}_C + d\overline{G}_D$$

That is, ΔG for the reaction is zero.

2. The state of reaction equilibrium is determined by the standard Gibbs free energy change for the reaction, ΔG°, via

$$\Delta G^\circ = -RT\ln K$$

where *K* is the equilibrium constant for the reaction given by the quotient of the activities of the reactants and products at reaction equilibrium; that is,

$$K = \frac{a_C^c a_D^d}{a_A^a a_B^b}$$

13.11 Summary

3. The Raoultian standard state of a thermodynamic component is the pure component in its stable state of existence at the temperature of interest. The Henrian standard state is obtained from consideration of Henry' s law, which, strictly being a limiting law obeyed by the solute *B* at infinite dilution, is expressed as

$$\frac{a_B}{X_B} \rightarrow k_B \quad \text{as} \quad X_B \rightarrow 0$$

where a_B is the activity of *B* in the solution with respect to the Raoultian standard state and *k* is the Henry' s law constant at the temperature *T*. Alternatively,

$$\frac{a_B}{X_B} \rightarrow \gamma_B^{\circ} \quad \text{as} \quad X_B \rightarrow 0$$

where $\gamma_B^{\circ}(= k_B)$ is the constant activity coefficient which quantifies the difference between Raoultian and Henrian solution behavior of *B*.

13.11 Summary

- If the solute obeys Henry’ s law over a finite range of composition, then, over this range,

$$a_B = \gamma_B^{\circ} X_B$$

- The Henrian standard state is obtained by extrapolating the Henry’ s law line to $X_B = 1$, and the activity of *B* in the Henrian standard state with respect to the Raoultian standard state having unit activity is

$$a_B = \gamma_B^{\circ}$$

- The activity of B in a solution with respect to the Henrian standard state having unit activity is given by

$$h_B = f_B X_B$$

where h_B is the Henrian activity and f_B is the Henrian activity coefficient. In the range of composition in which *B* obeys Henry’ s law, $f_B = 1$.

13.11 Summary

- ✓ The 1 wt% standard state is defined as

$$\frac{h_{B(1\text{ wt\%})}}{\text{wt\%}B} \to 1 \quad \text{as} \quad \text{wt\%}B \to 0$$

and is located at that point on the Henry' s law line which corresponds to a concentration of 1 wt% *B*.

- ✓ With respect to the 1 wt% standard state having unit activity, the activity of *B*, $h_{B(1\text{ wt\%})}$, is given by

$$h_{B(1\text{ wt\%})} = f_{B(1\text{ wt\%})}\text{wt\%}B$$

where $f_{B(1\text{ wt\%})}$ is the 1 wt% activity coefficient.

- ✓ The activities are related via

$$a_B = h_B \gamma_B^{\circ}$$

and

$$a_B = f_{B(1\text{ wt\%})} \cdot \text{wt\%}B \cdot \gamma_B^{\circ} \cdot \frac{\text{M}W_{\text{solvent}}}{100\text{M}W_B}$$

13.11 Summary

4. The Gibbs equilibrium phase rule is

$$F = C + 2 - \Phi$$

where:

C is the number of components in the system

F is the number of degrees of freedom available to the equilibrium involving Φ phases

With *R* independent reaction equilibria involving *N* species, the Gibbs equilibrium phase rule is

$$F = (N - R) + 2 - \Phi$$

where $C = N - R$

13.11 Summary

5. In a solution of solvent *A* and several dilutes solutes *B*, *C*, and *D*,

$$\ln f_B = \varepsilon_B^B X_B + \varepsilon_B^C X_C + \varepsilon_B^D X_D$$

where

$$\varepsilon_j^i = \left.\frac{\partial \ln f_j}{\partial X_i}\right|_{X_i \to 0}$$

is the interaction parameter of *i* on *j*.

The interaction parameters are related to one another by

$$\varepsilon_i^j = \varepsilon_j^i$$

13.11 Summary

If the concentrations of the dilute solutes are expressed in weight percent, then

$$\log f_B = e_B^B \text{wt\% } B + e_B^C \text{wt\% } C + e_B^D \text{wt\% } D$$

where

$$e_i^j = \left.\frac{\partial \log f_j}{\partial \text{ wt\% } i}\right|_{\text{wt\% } i \to 0}$$

The two interaction parameters of *i* and *j* are related by

$$e_j^i = \frac{\mathrm{MW}_j}{\mathrm{MW}_i} e_i^j$$

Also

$$\varepsilon_j^i X_i = 2.303 e_j^i \text{ wt\% } i$$